

University of Essex, MSc Artificial Intelligence

MSc Computing Project

**Economic Crisis Prediction: Early Warning Systems for
Banking Crises, Evaluating the Interpretability–Performance
Trade-off Under Structural Non-Stationarity**

Pavlos Papachristos • PP24589

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Abstract

Machine-learning early-warning systems (EWS) for banking crises often appear accurate under expanding-window testing, but later folds absorb information from crisis regimes unavailable to a real-time forecaster. This dissertation asks whether such models generalise when tested strictly on future crises, and whether failure reflects data volume, model choice or structural instability.

Methodologically, it evaluates logistic regression, Random Forest and an Explainable Boosting Machine (EBM) on macro-financial data for 18 advanced economies over 1989–2020, then examines FinBERT sentiment from central-bank speeches within the 2007 Global Financial Crisis cross-section.

Four findings emerge. First, extending the training history by roughly fifteen years does not resolve transfer to the GFC: both macro-only models remain below random in the only informative strict future test, so the additional historical sample provides no demonstrated prospective benefit. Second, EBM is useful diagnostically but not inherently robust: it achieves the strongest expanding-window performance yet collapses furthest in strict Split B, while its shape functions reveal marked training-window dependence. Third, central-bank sentiment produces no statistically distinguishable improvement once algorithm class is held fixed; the apparent gain when algorithm and features change together cannot be separated from noise in a single episode. Fourth, the combined evidence is most consistent with instability in the learned mapping between macroeconomic indicators and crisis risk across historical periods, rather than insufficient data or a simple model-selection problem.

Because only a few independent crisis episodes are available, genuine change in the data-generating process cannot be fully separated from estimation instability in sparse, correlated samples. Higher within-period accuracy or interpretability therefore does not establish prospective trustworthiness; EWS should be assessed with strict temporal splits and model-native stability diagnostics before operational use.

Chapter 1: Introduction

1.1 Background and Motivation

The failure to anticipate systemic banking crises carries severe economic consequences. The Global Financial Crisis of 2007–2009 exposed critical gaps in the analytical frameworks used by supervisory authorities to monitor financial system stability (Laeven and Valencia, 2020). Despite substantial research effort, the fundamental challenge remains: how to construct models that identify systemic risk build-up far enough in advance to permit effective policy intervention, and that remain reliable across structurally different crisis configurations. Systemic banking crises are among the most costly macroeconomic events a modern economy can experience: they are associated with large and persistent output losses, sharp increases in public debt as governments absorb financial-sector losses, and prolonged credit contractions that impede recovery (Reinhart and Rogoff, 2009; Laeven and Valencia, 2020). These costs create a strong policy rationale for early-warning systems (EWS) — statistical models that estimate the probability of a crisis over a medium-term horizon from observable macro-financial indicators, giving supervisory authorities enough lead time to deploy countercyclical tools before vulnerabilities crystallise. Since the 2007–2009 crisis this rationale has been institutionalised: the Basel III framework conditions the countercyclical capital buffer on indicators of excess credit, most prominently the credit-to-GDP gap (Borio and Lowe, 2002; Drehmann and Juselius, 2014), making the predictive quality of EWS indicators a matter of operational supervisory policy rather than of academic interest alone. The EWS literature has developed in successive phases. The first phase, from the late 1990s, established the binary-prediction paradigm through two complementary approaches: the signal-extraction method, which flags an indicator once it breaches a threshold calibrated to minimise a policymaker’s loss function (Kaminsky, Lizondo and Reinhart, 1998), and the multivariate logit framework, which estimates crisis probabilities directly from a panel of macro indicators (Demirgüç-Kunt and Detragiache, 1998). This phase converged on a robust empirical regularity — that credit booms are the single most reliable antecedent of systemic banking crises across long historical panels (Schularick and Taylor, 2012) — which continues to anchor the macro feature sets used in modern systems, including the one constructed here.

A second phase introduced machine-learning methods such as Random Forest and XGBoost, which improved discrimination metrics within standard expanding-window evaluation designs (Beutel, List and von Schweinitz, 2019; Alessi and Detken, 2018). That reliability deteriorates sharply under strict temporal validation, however: models trained exclusively on pre-crisis data consistently failed to meet the 0.75 AUROC benchmark needed for practically useful prediction.

A parallel development introduced textual data from central bank communications as a complementary signal (Hansen and McMahon, 2016; Correa et al., 2021). Sentiment extraction from BIS speeches using FinBERT (Yang, Uy and Huang, 2020) operationalises this channel systematically. Both developments raise a common question: under what conditions do the observed performance gains survive genuinely prospective testing, and what does the answer reveal about the EWS modelling framework? This dissertation addresses that question using an extended macro panel, a natively interpretable non-linear model, and strict temporal validation.

1.2 Research Problem

The research problem has two analytically distinct dimensions.

The first is methodological. Standard expanding-window cross-validation allows training folds to accumulate crisis-period data, which produces metrics that overstate prospective predictive performance; this problem is particularly severe in GFC-concentrated datasets.

The second is theoretical: the data-generating process that produces crisis outcomes may not be stationary across time. Models trained on one crisis type learn a mapping that does not generalise to structurally different episodes, regardless of training sample length or algorithm sophistication.

1.3 Aim, Objectives and Hypotheses

Research Aim: To investigate the extent to which ML-based EWS achieve genuinely prospective predictive performance, and to characterise the structural conditions under which their generalisation fails, using an interpretable non-linear benchmark that makes the failure mechanism directly visible.

Objectives:

- (O1) Extend the JST panel from 2003–2020 to 1989–2020 incorporating multiple pre-GFC episodes;
- (O2) evaluate a model sequence spanning M1 (logistic regression), M2 (Random Forest), M2.5 (EBM), and M3 (XGBoost + FinBERT), under consistent expanding-window and strict temporal validation;
- (O3) use EBM shape functions to provide direct evidence on whether the learned feature-risk mapping is regime-stable;
- (O4) assess the incremental contribution of FinBERT sentiment, controlling for the algorithm effect, and evaluate ensemble averaging under single-episode evaluation constraints;
- (O5) draw implications for EWS design and ML under distribution shift.

Hypotheses:

- H1 — M3 achieves higher AUROC/AUPRC than M2-matched within Stage 2.

H2 — at least one FinBERT feature ranks in the top five by SHAP attribution.
H3 — a quantifiable accuracy-interpretability trade-off exists across $M1 \rightarrow M2 \rightarrow M3$.
EBM extends H3 by testing whether interpretability and accuracy can be jointly achieved within the non-linear class.

1.4 Contribution

The dissertation makes four contributions.

Empirical: extending training history across diverse crisis episodes does not resolve prospective generalisation failure.

Methodological: quantifying the evaluation bias in expanding-window designs by comparing OOS metrics against strict temporal results.

Interpretability: EBM shape functions expose regime-specific misspecification directly, without post-hoc attribution approximation.

Theoretical: this provides convergent, mechanism-level evidence consistent with structural non-stationarity in the data-generating process rather than algorithm choice or attribution opacity, while sparse-sample estimation instability cannot be excluded; EBM's more severe prospective collapse relative to Random Forest in Split B is directionally consistent with this but, given the six-positive training constraint, is treated as corroborating rather than confirmatory.

1.5 MSc AI Requirement

This dissertation implements and critically evaluates three ML model classes: gradient-boosted trees (XGBoost), ensemble methods (Random Forest), and gradient-boosted GAMs (EBM), alongside a transformer-based NLP pipeline (FinBERT). The analysis of distribution shift and non-stationarity connects to fundamental questions in learning theory. The project applies expanding-window CV, bootstrap CIs, Granger causality, SHAP, and EBM shape function analysis, and uses exclusively publicly available secondary data with ethical review clearance (low risk, no human participants).

Code is available at <https://github.com/spike2025-art/eportfolio-uoe/tree/main> and the ePortfolio at <https://spike2025-art.github.io/eportfolio-uoe/>.

1.6 Dissertation Structure

Chapter 2 reviews the EWS, interpretability, central bank text, and ensemble literatures. Chapter 3 describes the research design, data, models, and evaluation framework. Chapter 4 presents results across all model specifications. Chapter 5 discusses the findings, limitations, practical recommendations and future research. Chapter 6 restates the principal conclusions and reflects on the contribution of the study. References and appendices follow.

Chapter 2: Literature Review

2.1 Early-Warning Systems for Banking Crises

The early-warning system literature originated with signal extraction approaches (Kaminsky, Lizondo and Reinhart, 1998) and logistic regression frameworks (Demirgüç-Kunt and Detragiache, 1998), establishing the binary prediction paradigm that subsequent machine learning work extended.

The landmark contribution of Schularick and Taylor (2012) demonstrated that credit booms are the single strongest predictor of systemic banking crises across a long historical panel, motivating the macro feature sets — credit aggregates, interest rate spreads, asset prices — standard in subsequent EWS research.

The operational translation of the credit-boom finding into a supervisory indicator is due to Borio and Lowe (2002), whose credit-to-GDP gap — the deviation of the credit-to-GDP ratio from a slow-moving trend — later became a reference conditioning variable for the countercyclical capital buffer under Basel III. Drehmann and Juselius (2014) argued that indicator evaluation cannot rest on discrimination statistics alone: a useful early-warning indicator must issue signals with enough lead time for macroprudential policy to act, remain stable as new observations arrive, and be interpretable to decision-makers. They formalise this through a policymaker’s loss function that weighs missed crises against false alarms, and show that the credit-to-GDP gap and debt-service ratios dominate on these criteria at horizons of one to three years. This lead-time and stability requirement — not raw AUROC — motivates both the strict temporal validation and the practically oriented 0.75 benchmark adopted in Chapter 4.

The JST Macrohistory Database (Jordà, Schularick and Taylor, 2017), the primary data source for this dissertation, assembles these series for eighteen advanced economies from 1870 to 2020, enabling evaluation across structurally diverse pre-GFC episodes.

Comparative studies confirm that no single indicator is sufficient. Babecky et al. (2013), pooling forty developed economies, find that domestic credit growth, rising asset prices and deteriorating current-account positions jointly precede banking crises, while individual indicators are unstable across specifications. Frankel and Saravelos (2012), testing the pre-2008 leading-indicator apparatus against the crisis itself, report that reserves and past real-exchange-rate appreciation were among the few variables with consistent predictive content, underscoring how much of the earlier framework failed on precisely the episode it was most expected to anticipate. Reinhart and Rogoff (2009) situate this fragility historically: recurring “this-time-is-different” reasoning reflects a genuine instability in which empirical relationships appear stable until the regime that produced them ends. That observation is the historical counterpart of the stationarity assumption this dissertation tests directly.

A critical limitation of the macro EWS framework is its implicit stationarity assumption: that the mapping between precursors and crisis outcomes is stable across time. The heterogeneity

of crisis mechanisms — domestic credit-boom collapses, contagion events, speculative attacks (Obstfeld, 1996), and exogenous shocks — implies that no single feature set is universally predictive.

Laeven and Valencia (2020) document significant cross-episode variation in systemic banking crisis characteristics, confirming that undifferentiated crisis labels conflate structurally distinct events. This non-stationarity assumption is the theoretical foundation for the strict temporal validation central to Chapter 4.

2.2 Machine Learning in Crisis Prediction

Alessi and Detken (2018) demonstrated that Random Forest models outperform logistic regression on standard EWS metrics. Beutel, List and von Schweinitz (2019) conducted the most comprehensive ML evaluation to date, finding consistent AUROC gains from machine learning over logistic regression within expanding-window designs — but documenting substantial deterioration under genuinely prospective evaluation. Their AUROC benchmark of 0.75 serves as the threshold for practically useful EWS performance in this dissertation’s temporal validation.

The closest methodological precedent to the present study is Bluwstein et al. (2023), who apply machine-learning classifiers to a long-run cross-country panel closely related to the JST database and use Shapley-value attribution to interpret them. They report that non-linear models — particularly tree ensembles and neural networks — materially improve on the logit, and that the slope of the yield curve carries predictive information beyond credit growth, with the two interacting non-linearly. Their interpretability analysis, however, is conducted primarily under cross-validated rather than strictly prospective evaluation and rests on post-hoc SHAP attribution. The present dissertation departs on both points: it evaluates under strict temporal splits that forbid any leakage of future crisis regimes into training, and it adopts a natively interpretable model whose shape functions are the fitted object itself rather than an approximation of it.

A wider body of work reaches broadly consistent conclusions while exposing the same evaluation caveat. Ward (2017) shows that classification-tree ensembles with many candidate predictors outperform conventional signalling models in sample, but cautions that predictor-selection instability inflates apparent performance. Tölö (2020) finds that recurrent neural networks exploit the temporal ordering of macro-financial series to improve crisis prediction, again chiefly under cross-validated designs. Fouliard, Howell and Rey (2021), by contrast, adopt an explicitly online-learning framework that updates only on information available at each date, and report that genuinely out-of-sample crisis prediction is considerably harder than pooled evaluation implies — a result that directly motivates the strict-future emphasis of this study. Collectively, the literature establishes that machine learning improves discrimination under standard testing, but leaves open whether those gains survive when the model is denied any sight of the future.

Under the class imbalance characteristic of crisis datasets (5–10 per cent positive rate), the choice of evaluation metric substantially affects conclusions. Davis and Goadrich (2006) demonstrate that AUPRC is more informative than AUROC under high class imbalance,

being more sensitive to improvements in crisis detection at the threshold values relevant for policy. Saito and Rehmsmeier (2015) confirm this formally. Both metrics are reported throughout; AUPRC is treated as primary in Stage 2.

2.3 Central Bank Communication and Text Analysis

The BIS Central Bank Speech Archive provides a consistent multilingual corpus spanning over two decades for the countries in the EWS panel. FinBERT (Yang, Uy and Huang, 2020), a BERT-based model fine-tuned on financial communications, provides sentence-level sentiment classification used in this dissertation to construct country-year sentiment features from 16,622 speeches.

Central bank language (hedged, indirect) presents a specific domain challenge: aggregate sentiment polarity captures less than the topic-specific dimensions of systemic risk concern identified by Ho, Moench and Steeley (2022), whose prevalence measures for credit risk discussion outperform polarity scores, consistent with Loughran and McDonald's (2011) broader finding that domain-specific lexicons outperform general sentiment tools on financial text. Both findings motivate the topic-specific extension proposed in Chapter 5.

The premise that central-bank language carries measurable macro-financial information is itself well established. Hansen and McMahon (2016), applying computational text analysis to Federal Open Market Committee communication, show that its forward-guidance component moves real and financial variables, distinguishing it from the mere disclosure of current conditions. In the financial-stability domain specifically, Correa et al. (2021) construct a sentiment measure from central banks' financial-stability reports and find that more negative sentiment predicts subsequent financial-sector stress, direct evidence that supervisory tone carries forward-looking risk information. Nyman et al. (2021) extend this to narrative dynamics, showing that shifts in the emotional content of financial text — rising excitement followed by anxiety — track the build-up and unwinding of systemic risk. These studies justify treating central-bank sentiment as a candidate signal, while their reliance on targeted, topic-aware measures rather than aggregate polarity anticipates the negative aggregate-polarity result reported in Chapter 4.

2.4 Interpretability in Machine Learning

The interpretability-accuracy trade-off is a central tension in applying machine learning to high-stakes domains. SHAP (Lundberg and Lee, 2017) provides post-hoc attribution for non-linear models using Shapley values, with exact computation for tree-based models via TreeSHAP. Despite wide adoption, Hooker and Mentch (2019) show permutation-based

methods including SHAP can produce misleading attributions under feature correlation — a concern in macro panels with strong multicollinearity — so SHAP attributions should be read as measures of model re-weighting, not causal economic importance.

Explainable Boosting Machines (Lou, Caruana, Gehrke and Hooker, 2013) represent an alternative: a gradient-boosted GAM that learns univariate shape functions and pairwise interactions, whose outputs are exact representations of what the model has learned rather than post-hoc approximations. Because shape functions are the model itself, instability in those functions across different training periods reflects a property of the fitted model rather than attribution approximation error and provides model-native evidence of instability in the fitted feature–risk mapping which, combined with additional evidence, may support an interpretation of data-generating-process non-stationarity — a property central to the regime instability analysis in Section 4.7.

The choice of a natively interpretable model over a post-hoc-explained black box is a substantive methodological position rather than a presentational preference. Rudin (2019) argues that for high-stakes decisions, post-hoc explanations of opaque models can be unfaithful to the underlying computation, and that interpretable models should be preferred wherever they are competitive, because an explanation that does not match the model can mislead precisely when scrutiny matters most. The generalised additive model with interactions on which the EBM is built has a documented capacity to expose data-quality and specification problems that remain hidden in ensembles: Caruana et al. (2015) show that such a model, applied to pneumonia-risk prediction, recovered a clinically implausible learned relationship that a black-box model of equal accuracy had silently absorbed. This diagnostic property — the ability to read the learned feature-risk mapping directly and judge its plausibility — is exactly what the present study exploits to examine regime stability, and is why the EBM, implemented through the InterpretML framework of Nori et al. (2019), is adopted as the central methodological instrument rather than as one classifier among several.

2.5 Ensemble Methods and Financial Contagion

Holopainen and Sarlin (2017) formalised the motivation for ensemble averaging in EWS settings: combining predictions across classifiers with different architectures consistently outperforms single-model selection under small-sample, crisis-concentrated evaluation, with the improvement largest where component disagreement peaks at crisis observations. Equal-weighted averaging is used here to avoid the circular weight-selection problem of deriving weights from OOS performance.

The cross-country financial linkages through which GFC stress propagated — documented via bank balance sheet network data by Cetorelli and Goldberg (2011) — are absent from the present feature set, which treats each country independently. This structural gap is consistent with the model’s weaker performance for contagion-driven crises (Germany, Belgium, Switzerland in 2007–2008) and is identified as the most ambitious future extension in Chapter 5.

Underlying all of these strands is a problem the machine-learning literature treats explicitly as dataset shift — the case in which the joint distribution of features and outcomes differs between training and deployment, so that a model optimised on historical data is no longer optimal, or even well calibrated, on future data (Quiñonero-Candela et al., 2009). Standard cross-validation implicitly assumes this shift is absent, which is precisely the assumption that fails when a small number of structurally distinct crisis episodes must each be predicted from the others. Framed this way, the recurring gap between cross-validated and strictly prospective crisis-prediction performance is not a defect of any single algorithm but a symptom of shift, and interpretable shape functions offer a way to observe that shift in the fitted model directly rather than infer it from aggregate metrics alone. This synthesis motivates the three specific gaps addressed below.

2.6 Research Gap

Three specific gaps motivate this dissertation.

First, the behaviour of EBM under strict temporal validation (and whether interpretable models are more or less susceptible to regime-specific overfitting) has not, to our knowledge, been systematically examined in the EWS literature.

Second, existing studies document prospective failure without providing a mechanism-level explanation; EBM shape functions enable direct observation of which feature-risk relationships break down across regime boundaries.

Third, sentiment-augmented EWS studies do not control for the algorithm effect — the possibility that performance gains attributed to sentiment reflect the algorithm switch rather than the sentiment signal itself. EBM on the same macro features as M2 provides a partial decomposition of this effect, holding the feature set fixed while the algorithm class changes.

Chapter 3: Methodology

This dissertation treats a fitted model not as evidence of a fixed underlying law connecting macroeconomic conditions to banking crises, but as a description of what held in the period it was fit on: crisis dynamics change with regulation, market structure, and policy regimes. This pragmatist stance — judging models by how well their summary survives testing outside their fit period, rather than by whether they recover some deeper truth — motivates the heavy reliance on expanding-window validation, strict temporal splits, and shape-function inspection throughout.

3.1 Research Design

This dissertation adopts a quantitative empirical research design using secondary panel data. The strategy is: a) to characterise the conditions under which ML-based EWS succeed and fail and b) to use an interpretable model class to make the failure mechanism directly visible. The framework is a two-stage augmented pipeline. Stage 1 evaluates macro-only models on a 1989–2020 extended panel across multiple pre-GFC crisis episodes. Stage 2 evaluates sentiment-augmented models on the 2003–2020 overlap window where FinBERT data is available.

3.2 Data Sources and Ethics

JST Macrohistory Database R6 (Jordà, Schularick and Taylor, 2017) covers 18 advanced economies from 1870–2020. After three-year lag construction, the usable Stage 1 panel is 569 country-year observations (1989–2020). Two minor data gaps were addressed by linear interpolation (Denmark government debt, 1997; Portugal house prices eliminated by lag construction).

BIS Central Bank Speeches: 16,622 speeches (2003–2020) across 18 countries were downloaded from the BIS archive, matched to country identifiers, and processed through FinBERT (ProsusAI/finbert) running approximately 12.4 hours on CPU. Speech-level sentiment scores were aggregated to country-year level by length-weighted mean. Full pipeline in notebooks NB03–NB04. A limitation not previously quantified: FinBERT is an English-language model, and BIS speech archive coverage is predominantly, but not exclusively, English, with translated-versus-original-language coverage varying by country (BIS, 2018) and speech-level language untagged during corpus construction — so the share of non-English or machine-translated text feeding FinBERT cannot be reconstructed from the retained pipeline outputs, and is recorded as an unresolved, rather than quantified, limitation.

Ethics: Publicly available secondary data. No human participants. Random seeds fixed at 42 throughout. Full code at <https://github.com/spike2025-art/eportfolio-uoe/tree/main>; ePortfolio at <https://spike2025-art.github.io/eportfolio-uoe/>.

Generative AI tools, including Claude (Anthropic) and ChatGPT, were used for the following purposes: (1) literature search support and summarisation of source material; (2) Python code review and debugging across the notebook pipeline; (3) review of draft chapters and identification of logical inconsistencies; and (4) iterative drafting and editing support for prose across Chapters 3–6, including sentence-level rewriting, consolidation of repeated content, and terminology consistency checks, all directed by, and reviewed and approved by, the author before inclusion. All empirical analysis, model design, statistical computation, result interpretation, and academic arguments are the original work of the author; generative AI tools did not generate, select, or interpret any result, statistic, or finding.

3.3 Target Variable and Features

The primary target `target_h2` marks a country-year positive if crisis onset falls within the following two years (37 positives at 6.5% in Stage 1; 26 at 8.15% in Stage 2). COVID-19 (2020) is excluded from the crisis chronology; the observation is retained in the held-out set.¹ As a sensitivity check, every prediction file underlying this dissertation was audited for scored (non-null) observations in 2019 and 2020: none exist, for any model, in either stage. All headline metrics are therefore structurally unaffected by the COVID-19 shock, since the pandemic period was never scored rather than merely excluded from crisis labelling. Macro features (41): Ten JST macroeconomic variables were used as inputs (total loans, mortgage credit, broad money, long-term interest rate, short-term interest rate, nominal house prices, government debt, CPI, current account balance, real GDP), plus term spread (long minus short rate) as an eleventh, expressed at lags 1–3 (33 level-lag features); five year-on-year growth rates at lag 1 (total loans, mortgage credit, broad money, house prices, real GDP); and three rolling credit trend features at lag 1 (three-year rolling mean and standard deviation of total loan growth, four-year rolling OLS slope of total loans). All features were demeaned using country-specific expanding means — the mean at each observation computed solely from prior years for that country, with cross-sectional means initialising early observations, preventing forward leakage — yielding 41 macro features total (33 level-lags + 5 growth rates + 3 rolling credit measures).

¹ 2020 rows are retained in the panel (16 of 18 countries; JPN drops out after 2018) with `target_h2 = 0`, consistent with JST's crisis chronology recording no onset through 2020. This label is unverifiable within JST R6 itself, since `target_h2` requires checking for crisis onset in the following two years (2021–2022), which fall outside the database's coverage. This has no bearing on any reported result: the expanding-window CV schedule for both stages ends its test folds at 2018 (§4.1), so no model ever produces a scored prediction for a 2019 or 2020 observation, regardless of the label's status.

Sentiment features (20): Seven FinBERT-derived sentiment measures were constructed per country-year: the three base FinBERT probabilities (P_{pos} , P_{neg} , P_{neutral}), net sentiment ($P_{\text{pos}} - P_{\text{neg}}$), and one-year first differences (ΔP_{pos} , ΔP_{neg} , $\Delta \text{net_sentiment}$). These were intended to be expressed at lags 1 and 2 only, yielding 14 sentiment features. A substring-match pattern in the NB05A feature-selection step (Cell 12) additionally captured related lag-3 and derived sentiment columns, so the M3 input set actually comprises 20 sentiment features rather than the intended 14. This is a feature-selection labelling discrepancy, not a data leakage or validity issue: the underlying computation, cross-validation folds, and the M3 results reported in Table 4.2 all correspond to the 61-feature set actually used (41 macro + 20 sentiment) and are internally consistent throughout Chapter 4. Table 3.1 and Appendix B, Table B.1 are corrected accordingly. These were combined with the 41 macro features to form the 61-feature M3 input set.

3.4 Model Specifications

The model sequence is chosen to span the methodological history of the early-warning literature while isolating the effect of each modelling decision. Logistic regression (M1) is retained as the canonical EWS baseline — the workhorse of the first-generation multivariate framework (Demirgüç-Kunt and Detragiache, 1998) — against which any gain from non-linearity must be measured. Random Forest (M2) represents the tree-ensemble family shown to improve on the logit in comparative EWS evaluations (Alessi and Detken, 2018; Ward, 2017) and serves as the black-box non-linear benchmark. The Explainable Boosting Machine (M2.5) is the study’s central instrument: a gradient-boosted generalised additive model with pairwise interactions (Lou et al., 2013; Nori et al., 2019) whose shape functions are the fitted model rather than a post-hoc approximation, which — following the case for intrinsically interpretable models in high-stakes decisions (Rudin, 2019; Caruana et al., 2015) — allows the feature–risk mapping to be inspected directly for regime stability, the diagnostic the black-box models cannot supply. XGBoost with FinBERT sentiment (M3) represents the sentiment-augmented frontier (Beutel, List and von Schweinitz, 2019; Yang, Uy and Huang, 2020) and is included so the sentiment channel can be tested against the macro-only models. Carrying the same macro feature set across M1, M2 and M2.5 before sentiment enters in M3 lets the analysis separate the algorithm effect from the feature effect.

Model	Algorithm	Features	Key settings	Rationale (source)
M1	Logistic Regression	19 (lag1 only)	L2 C=0.1, balanced weights	Demirgüç-Kunt and Detragiache (1998)
M1b	Logistic Regression	41 (lag1–3)	Bridge specification	Demirgüç-Kunt and Detragiache (1998)
M2	Random Forest	41 macro	500 trees, depth 5, balanced weights	Alessi and Detken (2018); Ward (2017)
M2.5 (EBM)	Explainable Boosting Machine	41 macro	5,000 rounds, 5 interactions, sample weights, no isotonic calibration	Lou et al. (2013); Rudin (2019); Nori et al. (2019)
M3	XGBoost + FinBERT	61 (macro + sent)	300 trees, depth 4, isotonic calibration (CalibratedClassifierCV)	Beutel et al. (2019); Yang, Uy and Huang (2020)

Model	Algorithm	Features	Key settings	Rationale (source)
M2-matched / M2.5-matched	RF / EBM	41 macro	Trained on 2003–2020 only	As M2 / M2.5

Table 3.1. Model specifications summary, with the primary source motivating each model class.

Hyperparameter and procedural settings. All model hyperparameters in Table 3.1 were fixed a priori; the training notebooks perform no grid or nested-cross-validation search, so the reported expanding-window metrics carry no selection bias from hyperparameter tuning. Class imbalance is handled throughout by balanced class weights (logistic regression, Random Forest), per-fold scale_pos_weight (XGBoost) and per-fold inverse-frequency sample weights (EBM), rather than by resampling. The chosen values are at or more conservative than the libraries’ defaults — stronger-than-default L2 penalisation for logistic regression ($C = 0.1$) and capped tree depths (Random Forest 5, XGBoost 4) — reflecting the small, highly imbalanced panel of roughly thirty-seven positive observations. For the EBM, the study’s central instrument, the interaction count (five) and learning rate (0.01) were set as deliberately conservative choices to limit overfitting on thin folds, and §4.7.4 (Appendix B.2) confirms empirically that results are insensitive to them: AUROC varies by only 0.0071 across five configurations, with every alternative’s AUPRC inside the baseline’s bootstrap interval. Linear interpolation of a single missing observation (Danish government debt, 1997) and expanding-window demeaning follow Chapter 3; bootstrap intervals use 1,000 resamples and are reported as indicative given panel dependence.

Notation. To avoid the naming inconsistency flagged in internal review, this dissertation uses a single fixed convention throughout: M1 (logistic regression), M1b (bridge logistic specification), M2 (Random Forest), M2.5 (EBM) (Explainable Boosting Machine), M3 (XGBoost + FinBERT), M4 (E1-EBM ensemble, following Holopainen and Sarlin (2017); §4.2), with the suffix “-matched” denoting the 2003–2020 sentiment-overlap-window variant (e.g. M2-matched, M2.5-matched). “EBM” and “M2.5 (EBM)” refer to the same model and are used interchangeably only where M2.5 has already been introduced in the immediate context; likewise “E1-EBM” and “M4” refer to the same ensemble and are used interchangeably once introduced, except where a reference is explicitly qualified as the original study’s own ensemble (§5.4), which is a distinct construct.

Class imbalance is handled by per-fold sample weights: $\text{fold_spw} = n_{\text{neg}} / n_{\text{pos}}$ applied within each expanding-window training fold. EBM uses the interpret package v0.7.8.

3.5 Evaluation Framework

Expanding-window CV: Stage 1 uses 27 evaluation folds (1992–2018) with a four-year burn-in; Stage 2 uses 12 folds (2007–2018). Years 2019–2020 are held out in both stages. OOS predictions are pooled for aggregate metric computation. Because the burn-in and hold-out years are not scored, Stage 1 aggregate metrics are computed on the 467 scored observations spanning 1993–2018, containing 28 positive labels rather than the 37 present in the full panel. These 28 scored labels correspond to 14 crisis onsets, of which 13 belong to the 2007–08 global financial crisis: 26 of the 28 scored positives (93 per cent) originate in that single

episode, the early-1990s cluster falling entirely within the burn-in window. Stage 1 is therefore multi-episode in its training history but predominantly single-episode in what it actually scores, a distinction that qualifies every Stage 1 comparison reported below. Strict temporal validation (the primary prospective test) trains exclusively on pre-crisis data: Split A (Pre-Nordic: 1989–1990 → 1991–1997) and Split B (Pre-GFC: 1989–2002 → 2003–2011). Split C (2003–2005 → 2006–2009) is infeasible due to zero training positives in the Stage 2 window.

LOCO validation holds out each country in turn across 15 countries with hold-out crisis events (AUS, CAN, NOR excluded — no hold-out positives).

Metrics: AUROC (benchmark 0.75; Beutel, List and von Schweinitz, 2019); AUPRC (primary under class imbalance in Stage 1, where multiple folds are available; Davis and Goadrich, 2006; Saito and Rehmsmeier, 2015); Brier Score (squared probability error); ECE (bin-level calibration). In Stage 2, AUPRC is computable in one cross-sectional fold only and measures country ranking within the 2007 GFC episode rather than temporal predictive performance; it is reported for comparability but interpreted accordingly. Bootstrap 95% CIs from 1,000 resamples are indicative only due to panel dependence.

3.6 EBM Shape Function Analysis and Ensemble Design

To test the stability of the learned mapping, EBM is fitted separately on two nested training windows — an early GFC-run-up window (2003–2006) and an expanded GFC-centred window (2003–2010, labelled GFC-inclusive) — and shape functions for term spread and credit growth at lag 1 are compared across training windows. Instability in shape functions reflects properties of the fitted model rather than attribution approximation error — eliminating post-hoc tools as a source of confound. Such instability is consistent with two non-mutually-exclusive explanations: instability in the underlying data-generating process across regimes, and regime-specific model variance arising from thin training samples in one or both sub-periods. Section 4.7.3 presents the observed instability and discusses the extent to which these explanations can be distinguished given the available data.

The primary ensemble uses equal-weighted averaging of fold-generated probabilities (performance-weighted schemes are circular when weights are derived from the same prediction pool): $E1\text{-EBM} = \text{avg}(\text{EBM}, M2\text{-matched})$. Model disagreement is quantified as the mean absolute difference between component probabilities, computed separately at crisis and non-crisis observations.

The full notebook pipeline (Notebooks 01–11) is documented in Appendix B.

Chapter 4: Data Presentation & Analysis

Four models structure this chapter, each isolating one dimension of the comparison: M1 (logistic regression) is the interpretable linear floor; M2 (Random Forest) keeps the same macro features but adds non-linearity, isolating an algorithm effect; M2.5 (EBM) retains that non-linear capacity but constrains it to additive, natively inspectable shape functions — the model this dissertation leans on most, since its learned structure can be read directly rather than approximated post hoc; M3 (XGBoost with FinBERT sentiment) changes both algorithm and feature set at once, which is why the matched pairwise decomposition in §4.2.1 exists, to separate those two effects rather than let a single AUPRC number speak for both.

4.1 Dataset and Experimental Setup

This chapter's purpose is not to establish predictive superiority of sentiment-augmented models in any general sense, but to characterise precisely where apparent performance gains arise, under what conditions they survive, and why they warrant caution under severe data constraints. This framing — and the evaluation constraints set out below — governs all results in this chapter, including EBM, regardless of algorithmic sophistication or interpretability.

The analysis adopts a two-stage design that extends the original 2003–2020 macro panel backwards to 1985, using the JST Macrohistory Database R6 (Jordà, Schularick and Taylor, 2017). Stage 1 covers the macro-only models (M1, M1b, M2 and M2.5) on the full extended panel. After feature construction eliminates the warm-up rows, the usable dataset comprises 569 country-year observations from 18 advanced economies over 1989–2020. The binding constraint is the lag-1 variant of the four-year rolling credit features (`tloans_slope4_lag1` spans $t-4$ to $t-1$), so with raw coverage from 1985 the first computable year is 1989 after listwise deletion on the macro feature set. The primary prediction target, `target_h2`, marks a country-year positive if a systemic banking crisis onset falls within the following two years, yielding 37 positive observations against 532 negative and a base rate of 6.5 per cent.

The extension period (1989–2002) contributes 11 positives from the Nordic banking crisis cluster and further country-events; the 2003–2020 period contributes the remaining 26. These 37 positive country-year labels correspond to 19 distinct crisis onsets, since the two-year target horizon assigns up to two labels to each onset. The onsets in turn cluster into three systemic episodes: the early-1990s European and Nordic cluster (six onsets), Japan 1997 (one onset), and the 2007–08 global financial crisis (13 onsets, 65 per cent of the total). The effective number of independent crisis regimes available for learning is therefore three rather than thirty-seven, a constraint that conditions the interpretation of every subsequent result and motivates the strict temporal splits reported in §4.5. Stage 2 covers M2-matched, M2.5-

matched and M3 on the 2003–2020 overlap window where FinBERT sentiment data are available, with 26 positives and a base rate of 8.15 per cent. Two minor data gaps were addressed by linear interpolation (Denmark government debt, 1997) or eliminated by lag construction (Portugal house prices, 1985–1987).

Macro features are demeaned using expanding country-specific means consistent with the procedure in Chapter 3; model specifications are as per Table 3.1. The key addition is M2.5 (EBM) — a gradient-boosted generalised additive model (Lou et al., 2013) that learns univariate shape functions and up to five pairwise interaction terms, producing a natively interpretable model whose feature contributions can be read directly without post-hoc attribution. Expanding-window CV for Stage 1 uses 27 evaluation folds (1992–2018); Stage 2 uses 12 folds (2007–2018); 2019–2020 are held out in both stages.

Two constraints govern all interpretation. First, Stage 2 AUPRC is computable in one fold only (crisis events concentrate in 2007), so Stage 2 results constitute a cross-sectional case study of the 2007 GFC — measuring in-crisis discrimination across 18 countries, not temporal predictive validity — and are labelled the 2007 GFC Cross-Sectional Case Study throughout, with Stage 2 retained as a pipeline label only. Second, in the only informative prospective split both models fall substantially below the 0.75 AUROC benchmark (Section 4.5), so all expanding-window metrics are upper bounds on prospective performance.

4.2 Primary Model Comparison

Table 4.1 presents Stage 1 discrimination and probability metrics for M1, M1b, M2 and M2.5 (EBM) on the extended panel. Table 4.2 presents 2007 GFC Cross-Sectional Case Study metrics for M2-matched, M2.5-matched, M3 and M4 (E1-EBM). Bootstrap 95 per cent confidence intervals from 1,000 resamples are indicative only.

Table 4.1. Stage 1 — Model comparison on extended macro panel (H2 target, 1989–2020)

Model	Features	AUROC	95% CI	AUPRC	AUPRC/Base	Brier
M1: Logistic Regression (macro lag1)	19	0.606	[0.477, 0.739]	0.139	2.3×	0.157
M1b: LR bridge (macro lag1–3)	41	0.646	—	0.145	2.4×	0.136
M2: Random Forest (macro lag1–3)	41	0.636	[0.484, 0.781]	0.224	3.7×	0.060
M2.5 (EBM, macro lag1–3)	41	0.753	[0.649, 0.855]	0.273	4.5×	0.064

Table 4.1. Stage 1 evaluation metrics on H2 (2-year) target. Full panel: 37 positive country-year labels among 569 observations (6.5%); metrics are computed on the 467 scored observations containing 28 positive labels, a scored prevalence of 6.0% which is the no-skill baseline for the AUPRC/Base column. CI: 1,000 bootstrap resamples; indicative only given panel dependence and the concentration of 93 per cent of scored positives in the GFC episode. Stage 1 covers 1989–2020; 27 evaluation folds (see evaluation scope, §4.1). M2.5 uses identical features and CV design as M2; the magnitude of EBM’s apparent advantage is subject to the same regime-specific evaluation bias that inflates all expanding-window figures.

Table 4.2. 2007 GFC Cross-Sectional Case Study — In-episode country discrimination, sentiment overlap period (H2 target, 2003–2020)

Model	Features	AUROC	95% CI	AUPRC	AUPRC/Base	Brier
M2-matched: RF (macro lag1–3)	41	0.939	[0.897, 0.970]	0.343	6.7×	0.060
M2.5-matched (EBM, macro lag1–3)	41	0.960	[0.927, 0.988]	0.532	10.4×	0.067
M3: XGBoost + FinBERT (macro + sent)	61	0.932	[0.875, 0.979]	0.411	8.0×	0.061
M4: E1-EBM: Ensemble avg(EBM, M2-matched)	—	0.9545	-	0.4329	8.5×	0.0557

2007 GFC Cross-Sectional Case Study metrics on H2 (2-year) target. Full Stage 2 panel: 26 positive country-year labels among 319 observations (8.15%); metrics are computed on the 215 scored observations containing 11 positive labels, a scored prevalence of 5.1% which is the no-skill baseline for the AUPRC/Base column (see Table 4.6). AUPRC is computable in one fold only, measuring how well each model ranks the 18 countries by crisis probability within the single known 2007 episode; it does not measure temporal predictive performance across multiple out-of-sample periods. Bootstrap 95% CI for AUPRC difference (M3 – M2-matched): $[-0.130, +0.312]$ — this interval spans zero; the directional difference cannot be distinguished from noise (see evaluation scope, §4.1). E1-EBM ensemble (avg(EBM, M2-matched)) is reported without bootstrap CI; AUPRC of 0.4329 falls below EBM standalone (0.532), reflecting probability dilution from averaging with M2-matched.

Figure 4.1 presents ROC and PR curves for all models across both stages. The Stage 1 curves (solid) confirm that M2.5 appears to achieve a higher true positive rate at low false positive rates than M2 on the extended panel; Stage 2 dashed curves cluster at high AUC values for all models, reflecting GFC concentration. The PR curves show the wider dispersion of Stage 1 results — where the longer historical training window produces more modest estimates, although the scored positives remain overwhelmingly GFC-concentrated — against the compressed Stage 2 figures.

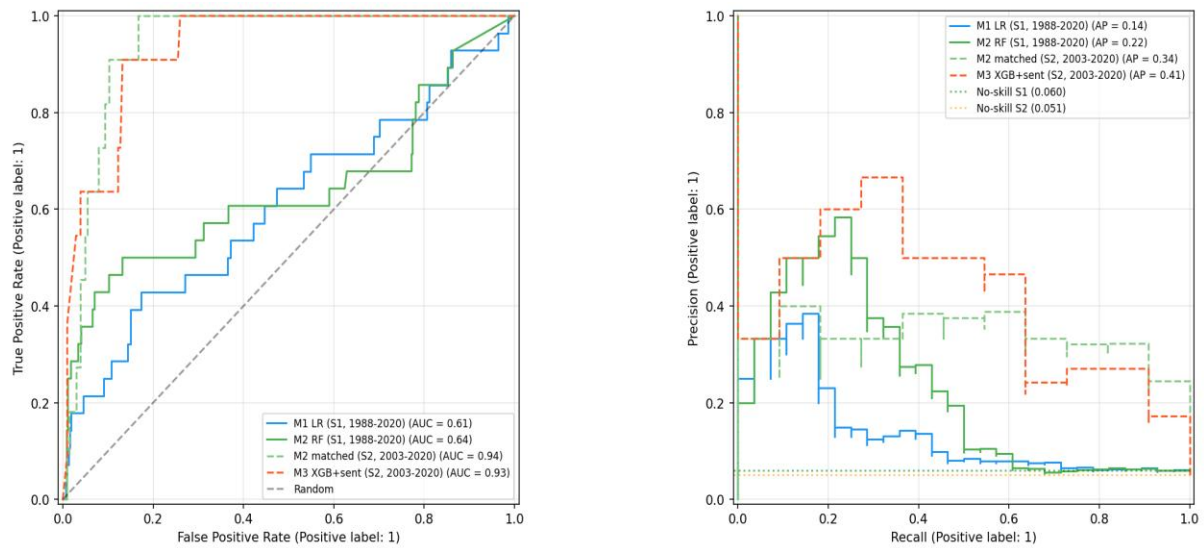


Figure 4.1. ROC and PR curves for all models across Stage 1 (solid lines) and Stage 2 (dashed lines). Stage 1 curves: 467 scored observations, 28 positive labels, 1993–2018 (full analytical panel 569/37, 1989–2020). Stage 2 curves: 215 scored observations, 11 positive labels (full Stage 2 panel 319/26, 2003–2020). No-skill baselines shown as dotted horizontal lines. All metrics are upper bounds on prospective performance. The apparent Stage 2 superiority of M2.5-matched (orange dashed) is conditional on the single-episode GFC evaluation setting.

Stage 1 results are far more modest than the original analysis: M2 collapses from AUROC 0.947 and AUPRC 0.561 in the original to the Table 4.1 figures, reflecting the far more demanding 27-fold expanding-window design applied to an extended training history rather than a like-for-like comparison. M2.5 (EBM) posts the highest Stage 1 point estimates on identical features and CV design. A paired DeLong test confirms the 0.117 AUROC gap is significant ($z = -3.35$, $p < 0.001$, 95% CI $[-0.186, -0.049]$); as established in Section 4.1, this describes point estimates within the same regime-specific window rather than prospective validity. Because pooled panel predictions violate the DeLong independence assumption, the comparison was repeated by cluster bootstrap (5,000 replicates): resampling the 18 countries returns a mean gap of +0.117, 95% interval $[+0.068, +0.175]$ ($p = 0.0002$), and a leave-one-country-out jackknife moves it only between +0.101 and +0.127 (sd 0.007), robust to cross-sectional dependence and not driven by any single country. Resampling by year widens the interval to $[-0.011, +0.248]$ ($p = 0.095$), and excluding the 2006 label-year reduces the gap to +0.019. The advantage is a stable cross-country regularity within the evaluated period, but the evidence that it persists in a structurally different regime is weak, since 93 per cent of scored positives originate in the GFC; it is reported as a within-regime finding, not a general property of interpretable models.

Paired DeLong tests on the 2007 cross-sectional AUROC values agree: M2-matched (0.939) versus EBM-matched (0.960) gives $z = -1.66$, $p = 0.097$, 95% CI $[-0.047, 0.004]$; EBM-matched versus M3 (0.932) gives $z = 1.39$, $p = 0.164$, 95% CI $[-0.012, 0.068]$. Neither is

distinguishable from zero: with 11 positives in a single episode, the cross-section lacks the power to separate these three models on ranking alone.

As Table 4.2 shows, M3's 6.8-percentage-point AUPRC advantage over M2-matched does not survive the bootstrap CI test $[-0.130, +0.312]$, which spans zero; the comparison also confounds algorithm and feature set simultaneously. The more informative test is M2.5-matched (EBM, macro-only, AUPRC 0.532) versus M3 (XGBoost, macro plus sentiment, AUPRC 0.411): the macro-only interpretable model outperforms the sentiment-augmented one, indicating that 2007 cross-sectional differences are at least as sensitive to model class as to the sentiment extension.

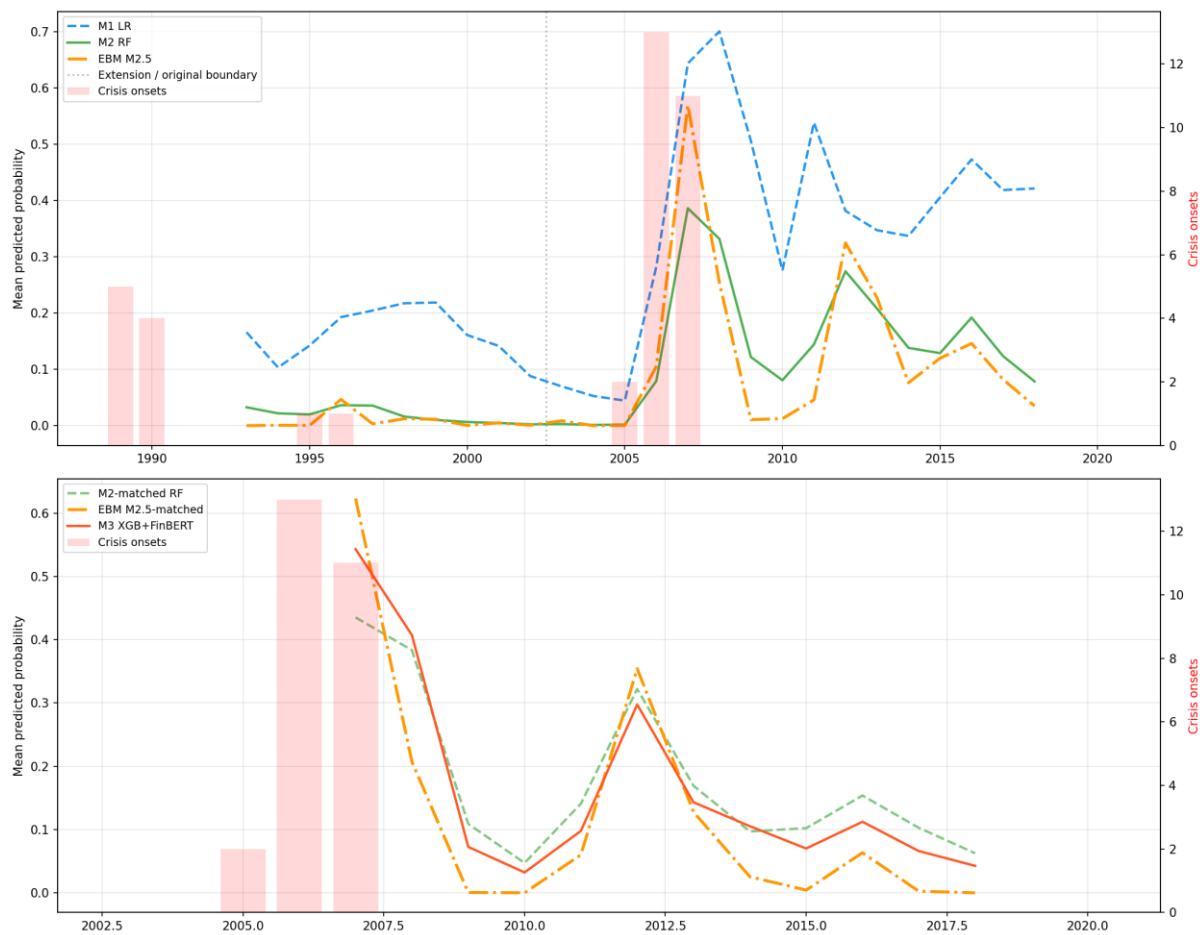


Figure 4.2. Out-of-sample crisis probability over time for all models. Stage 1 (top): M1 LR (blue dashed), M2 RF (green solid), M2.5 (EBM) (orange dash-dot), 1989–2020. Stage 2 (bottom): M2-matched RF (green dashed), M2.5-matched (orange dash-dot), M3 XGB+FinBERT (red solid), 2003–2020. Red bars = crisis onset years. Dotted vertical line = extension/original period boundary.

H1 verdict: not confirmed (directional only, CI spans zero; algorithm class explains at least as much of the cross-sectional ranking difference as the sentiment extension). Full assessment in Table 4.7, §4.11.

A model averaging extension, designated M4 (E1-EBM), is introduced following Holopainen and Sarlin (2017) as the equal-weighted average of EBM (M2.5-matched) and M2-matched fold-generated probabilities. M4 achieves AUROC 0.9545, AUPRC 0.4329, and Brier Score 0.0557 within the 2007 cross-sectional evaluation (Table 4.2).

4.2.1 Isolating the Algorithm and Sentiment Effects

The M2.5-matched versus M3 comparison confounds algorithm class (EBM versus XGBoost) and feature set (macro-only versus macro plus sentiment). A matched pairwise decomposition of algorithm and sentiment effects compares the relevant algorithm–feature combinations on the 2007 GFC cross-sectional panel; all carry the §4.2 scope limitations (single fold, directional only, no CI for the algorithm effect, neither effect distinguishable from noise). The results are AUPRC 0.3428 (RF, macro-only), 0.5320 with Brier 0.0672 (EBM, macro-only), 0.4537 with Brier 0.0909 (EBM, macro plus sentiment), and 0.4110 with Brier 0.0608 (XGBoost, macro plus sentiment).

Two paired comparisons follow, each varying one design element at a time. Holding the feature set fixed at 61 features and varying only the algorithm (EBM versus XGBoost), AUPRC rises by +0.043 in EBM's favour: a directional algorithm-class difference within this episode. Holding the algorithm fixed at EBM and varying only the feature set (macro-only versus macro plus sentiment), AUPRC falls by −0.078: a directional feature-set difference in the opposite direction, consistent with sentiment reducing EBM's within-episode discrimination once algorithm class is held constant.

Three qualifications apply. First, a bootstrap 95% CI for the isolated EBM feature effect (1,000 resamples of the 215 paired inner-merged out-of-sample predictions from NB05B and NB12) gives [−0.292, +0.187] for the −0.078 difference — spanning zero by a wide margin and confirming its directional-only status. Second, the decomposition is not orthogonal (Table 4.1): EBM and XGBoost differ in regularisation, learning dynamics, and interaction handling beyond algorithm class alone. Third, the comparison is non-stationary across cells — EBM macro-only uses 41 features, EBM macro+sentiment 61 — so feature count is itself a confound within the "feature effect."

Subject to these constraints, the decomposition is directionally consistent with algorithm class explaining at least as much of the cross-sectional ranking difference as the sentiment extension, with the isolated sentiment contribution negative rather than positive within this episode. This is a descriptive observation from a single cross-section, not a formal demonstration; it is corroborated by §4.9, which finds no sentiment Granger or panel-logistic signal surviving multiple-testing correction.

4.3 Exploratory Decomposition of Model Changes

Table 4.1 reports the full M1→M1b→M2→M2.5 sequence. Holding the feature set constant (41 macro features), the non-linearity step (M1b→M2) produces the largest linear-step AUPRC gain (+0.079), while the algorithm step (M2→M2.5) shows the largest single-step AUROC gain (+0.117) — a natively interpretable model achieving higher point estimates than its black-box counterpart on identical features. Whether this survives prospective evaluation is answered in Section 4.5: it does not.

H3 verdict — original (pre-specified): partially supported, but not in monotonic form. The linear-to-black-box transition shows an accuracy–interpretability trade-off, whereas the EBM demonstrates that native interpretability and strong within-regime point estimates can coexist. Exploratory extension: the EBM's larger Split B collapse is directionally consistent with flexibility-related regime overfitting, but is not a confirmatory H3 result, and neither interpretability nor algorithm choice resolves the regime instability problem (Section 4.5). Full assessment in Table 4.7, §4.11.

4.4 Multi-Horizon Analysis

Alternative prediction horizons (H1, H3) cannot be evaluated on the Stage 2 dataset: the held-out period contains insufficient positive cases under all three horizon definitions, a structural consequence of crisis concentration in 2007. The original multi-horizon findings are retained only as preliminary, GFC-specific observations.

4.5 Robustness: Temporal Validation and Leave-One-Country-Out

4.5.1 Strict Temporal Validation

The strict temporal validation is the most demanding robustness test here: its results establish the upper bound on prospective performance for any model trained on this panel, and should be read against all expanding-window metrics accordingly. Table 4.4 reports both M2 (Random Forest) and M2.5 (EBM).

Table 4.4. Strict temporal validation — M2 RF and M2.5 EBM (macro lag1–3)

Split	Train	Test	Train pos.	Test pos.	M2 RF AUROC	EBM AUROC	Benchmark
A: Pre-Nordic	1989–1990	1991–1997	5	1	0.572	0.752	0.75
B: Pre-GFC extended	1989–2002	2003–2011	6	13	0.481	0.333	0.75
C: Pre-GFC short	2003–2005	2006–2009	0	—	—	—	—

Table 4.4. Strict temporal validation. Literature AUROC benchmark = 0.75 (Beutel, List and von Schweinitz, 2019). Split C skipped: zero training positives. Split A: one positive test case — not a distributional estimate, reported for completeness only. Train pos. and Test pos. columns count distinct crisis onsets, not positive country-year labels; each onset generates up to two labels under the two-year horizon. Split B is the only split with sufficient test cases (13) to provide distributional information on prospective performance; however, with only six pre-GFC training positives, the training set is too thin to yield reliable model estimates, and results must be treated as indicative rather than conclusive. Neither model meets the 0.75 benchmark in Split B. The difference in AUROC between EBM (0.333) and RF (0.481) is directionally consistent with the regime-specific overfitting hypothesis but cannot be statistically confirmed from a single constrained split of this size.

The EBM–RF gap (0.148 points, no valid CI) is directionally consistent with higher representational flexibility producing more severe regime-specific overfitting under structural breaks, but rests on a single split with only six pre-GFC positives, so it is read as directional rather than confirmed; §4.7's shape-function evidence provides the more direct basis for the overfitting argument. Neither extending training history nor changing algorithm class produces a model that generalises prospectively to the GFC.

4.5.2 Leave-One-Country-Out Validation

Figure 4.3 presents the LOCO AUROC comparison for M2 RF and M2.5 EBM across 15 countries with hold-out crisis events. Three countries with no hold-out crisis events (Australia, Canada, Norway) are excluded from the analysis as AUROC is undefined without

positive cases.

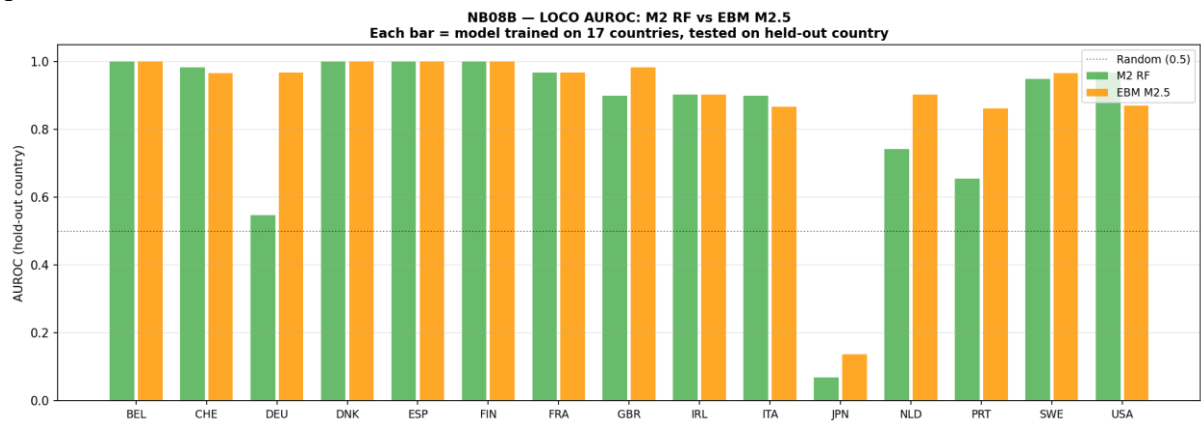


Figure 4.3. Leave-one-country-out AUROC: M2 RF (green) vs M2.5 (EBM) (orange) for each held-out country, Stage 1 (1989–2020). Each bar represents a model trained on 17 countries, evaluated on the held-out country. Dotted line = random classifier (0.5). AUROC = 1.000 entries reflect trivial single-event ranking.

Table 4.5. Leave-one-country-out summary — M2 RF vs M2.5 EBM, Stage 1

Country	Hold-out events	RF AUROC	EBM AUROC
BEL	1	1.000	1.000
CHE	2	0.983	0.967
DEU	1	0.548	0.968
DNK	1	1.000	1.000
ESP	1	1.000	1.000
FIN	1	1.000	1.000
FRA	1	0.968	0.968
GBR	2	0.900	0.983
IRL	1	0.903	0.903
ITA	2	0.900	0.867
JPN	1	0.069	0.138
NLD	1	0.742	0.903
PRT	1	0.655	0.862
SWE	2	0.950	0.967
USA	1	0.968	0.871
Mean (std)		0.839 (0.254)	0.893 (0.215)

Table 4.5 Leave-one-country-out summary — M2 RF vs M2.5 EBM, Stage 1. AUROC = 1.000 entries reflect trivial single-event ranking. Japan (both models near-random:

$RF=0.069$, $EBM=0.138$) is consistent with asset-deflation crisis dynamics being poorly represented in the cross-country training distribution. Mean figures are sensitive to single-event countries in both directions rather than uniformly inflated: the median AUROC (EBM 0.9667, RF 0.9500) and the mean restricted to the four countries with more than one hold-out crisis event (EBM 0.9458, RF 0.9333) are both higher than the reported means, indicating single-event volatility pulls the mean down as often as it inflates it; the cross-sectional pattern reflects sample-specific fitting behaviour rather than stable cross-country generalisation.

EBM shows a higher mean AUROC (0.893 versus 0.839) and lower standard deviation (0.215 versus 0.254) than RF, and the ordering holds under the more robust median and multi-event statistics above, so the advantage is not a single-event artefact. Given the prevalence of single-event hold-outs, country-level differences should still be read cautiously, since AUROC there ranks one positive against all negatives rather than measuring distributional discrimination. The pattern is consistent with the Stage 1 result and does not alter the prospective generalisation failure established in Section 4.5.1.

4.6 SHAP Feature Attribution

4.6.1 Aggregate Attribution by Lag Group

Figure 4.4 presents SHAP attribution for M3 in Stage 2 by lag group: the macro share is 73.8 per cent and FinBERT sentiment 26.2 per cent.

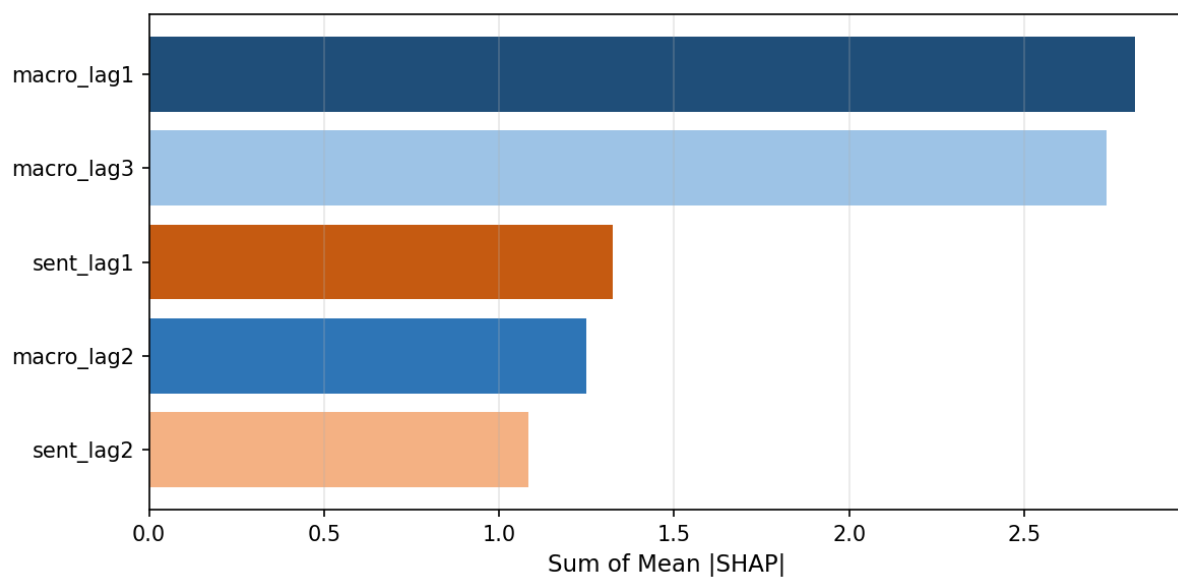


Figure 4.4. M3 SHAP signal by lag horizon (Stage 2, 2003–2020). Dark blue = macro lag1 (2.816); light blue = macro lag3 (2.735); medium blue = macro lag2 (1.249); orange tones = FinBERT sentiment lag1 (1.325) and lag2 (1.083). Macro features account for 73.8% of total attribution; sentiment 26.2%.

Within the macro group, lag-one attribution marginally leads lag-three and substantially leads lag-two; the lag-three signal indicates M3 draws on macro conditions up to three years before the crisis window, consistent with the slow build-up dynamics in §4.9. The sentiment share confirms FinBERT features contribute non-trivially within the GFC episode, but reflects internal weighting in a single episode rather than a stable or causal signal generalisable beyond 2007.

4.6.2 Individual Feature Rankings

The top-20 SHAP ranking places the three-year lagged term spread (term_spread_lag3_dm) first by a substantial margin, followed by the one-year lagged term spread and the four-year credit-growth slope; five sentiment features appear in the top 20, reflecting the GFC-concentrated sample.

H2 verdict: not supported in its strict formulation — no sentiment feature ranks in the top five by SHAP or EBM native importance. Full assessment in Table 4.7, §4.11.

4.6.3 Convergence with EBM Native Importance

Figure 4.5 presents SHAP feature rankings for M3 alongside EBM native feature importance for M2.5, derived without post-hoc attribution.

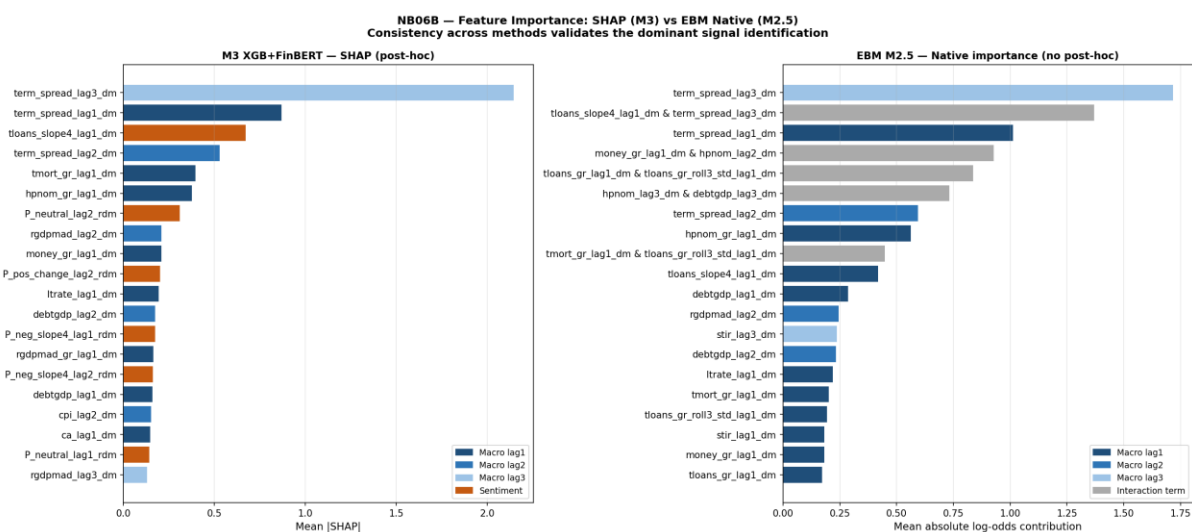


Figure 4.5 Feature importance comparison: M3 SHAP (left, post-hoc) vs M2.5 (EBM) native importance (right, no post-hoc). Dark blue = lag1; medium blue = lag2; light blue = lag3;

orange = sentiment (SHAP only); grey = interaction term (EBM only). Both methods independently identify term_spread_lag3_dm as the dominant feature.

SHAP ranks term_spread_lag3_dm highest by mean |SHAP|; EBM assigns it native importance 1.717 — the top feature by both methods, a methodological triangulation showing the finding is not an artefact of the model class or the attribution tool, though the agreement reflects shared sample-specific correlation within the GFC episode rather than causal invariance across regimes. EBM additionally surfaces a credit-slope / term-spread interaction as the second-ranked feature (§4.7.2).

4.7 EBM Interpretability Benchmark

4.7.1 Motivation and Design

M2.5 (EBM) is introduced between M2 and M3 because M2 and M3's feature-risk relationships are accessible only through post-hoc attribution such as SHAP (Lundberg and Lee, 2017), which is approximate, non-causal, and sensitive to distributional shift (Hooker and Mentch, 2019). EBM's shape functions are the model itself, so any instability observed in §4.7.3 cannot be attributed to approximation error — that confound is eliminated by construction, leaving only instability in the fitted model, whose two candidate explanations (DGP non-stationarity or regime-specific variance from thin samples) §4.7.3 assesses in full.

4.7.2 Feature Importance

Figure 4.6 presents EBM's global feature importance for the Stage 2 full-data fit.

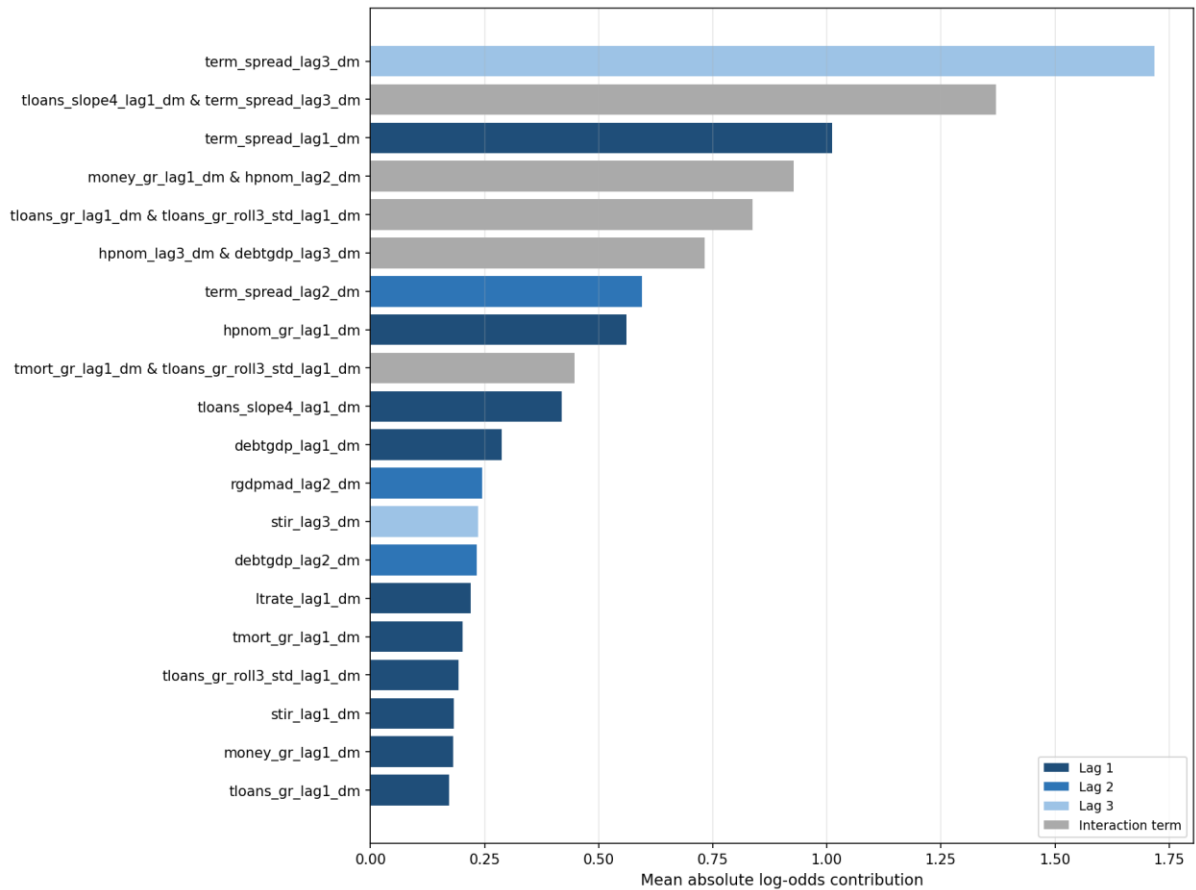


Figure 4.6. M2.5 (EBM) global feature importance (Stage 2, 2003–2020). Dark blue = lag1; medium blue = lag2; light blue = lag3; grey = interaction term. Term spread at lag 3 (importance 1.717) leads substantially. Term spread appears at all three lag horizons in the top seven features. Signal is concentrated in a small number of economically coherent macro predictors; no sentiment feature appears in the top 20.

EBM's top five main-effect features are entirely macro variables dominated by the term spread and credit trend: term spread at lag three (importance 1.717) leads by a margin of 0.705 over lag one (1.012), with lag two (0.595), nominal house price growth at lag one (0.562), and the four-year credit slope at lag one (0.419) completing the top five. The interaction between the four-year credit slope and the three-year lagged term spread ranks second overall among all features including interactions (importance 1.370), indicating that EBM identifies a joint credit-term spread signal beyond either variable's individual contribution.

These importances are derived from a full-data fit on the Stage 2 panel and reflect in-sample feature weighting within the GFC-dominated training distribution.

4.7.3 Shape Function Analysis: Training-Window Instability

The early GFC-run-up EBM is trained on the 2003–2006 sub-period: 72 country-year observations (18 countries $\times$ 4 years), of which 15 are crisis-positive under the two-year-

forward target_h2 indicator — 2 in 2005 and 13 in 2006. These 15 observations are not 15 independent crisis signals: because target_h2 looks two years ahead, both years fall within the immediate run-up to the single 2007–2008 GFC episode, so the positives are highly correlated realisations of one systemic shock rather than historically distinct crisis events. The GFC-inclusive EBM is trained on 2003–2010, 144 observations, capturing all 26 crisis positives in the full panel (2 in 2005, 13 in 2006, 11 in 2007) — none occur after 2007. Shape function estimates under EBM are sensitive to both the number and the effective independence of training positives, particularly in extreme feature-value bins; this context governs interpretation of the instability analysis below.

Figure 4.7 presents EBM shape functions for term spread and total credit across all three lag horizons, estimated separately on an early GFC-run-up window (2003–2006) and a GFC-inclusive window (2003–2010). This is the centrepiece analysis of the EBM contribution.

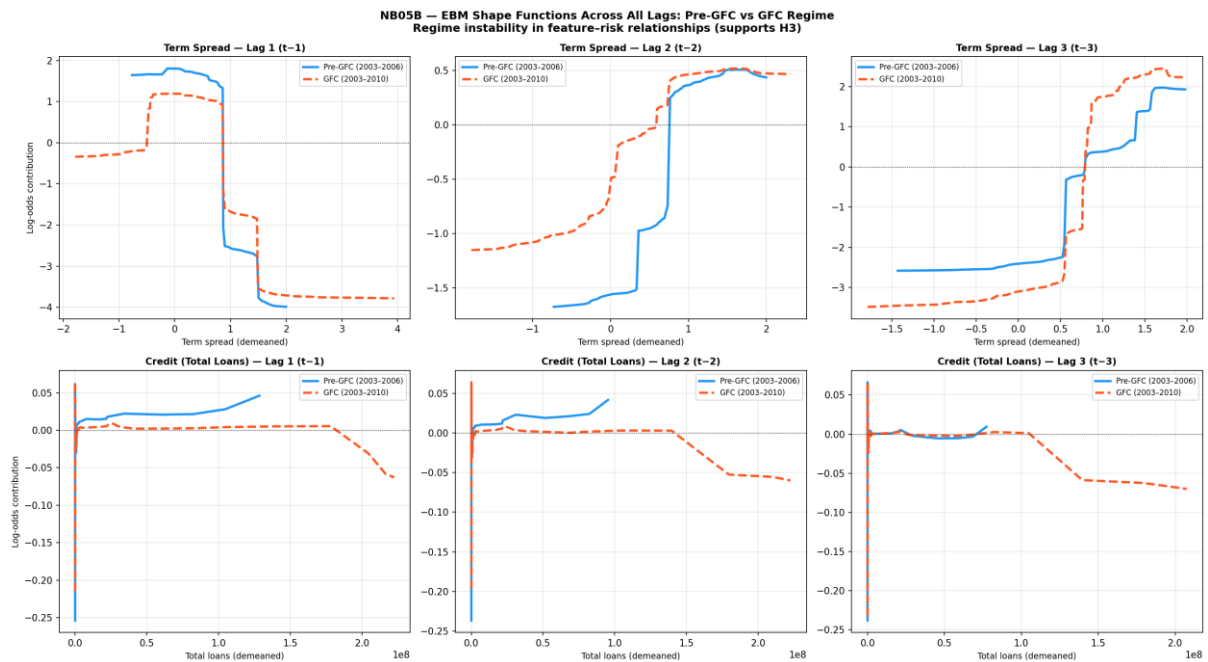


Figure 4.7. EBM shape functions across all lags: Early GFC-run-up (blue solid, 2003–2006) vs GFC-inclusive (red dashed, 2003–2010). Top row: term spread at lags 1, 2 and 3. Bottom row: total credit at lags 1, 2 and 3. Y-axis: log-odds contribution to crisis probability. Divergence between curves indicates training-window instability in the learned feature-risk mapping.

The lag-3 term spread panel (top right) shows qualitatively distinct shapes — not merely scaled versions — and the pre-GFC and GFC-inclusive curves cross at moderate positive spread values, meaning the same feature value is associated with opposite-sign risk contributions across the two training windows. Because shape functions are the model itself, this divergence cannot be attributed to post-hoc approximation error.

Two non-exclusive explanations warrant consideration, and the data cannot fully separate them. The first is DGP non-stationarity: the mapping from term spread and credit growth to

crisis risk genuinely differs between regimes, consistent with the Chow-type structural break for credit growth (§4.9.7: LR = 14.242, $p = 0.0066$) and the panel logistic sign reversal across lags (§4.9.5: lag-1 coefficient -0.646 against lag-3 coefficient $+0.630$). The second is regime-specific model variance: the pre-GFC sub-period's limited, temporally clustered positives (disclosed above) subject EBM's shape estimates to high variance regardless of DGP stationarity, so the instability could partly reflect estimation noise on sparse, correlated data rather than a structural DGP property.

The sign reversal's magnitude is large enough to be economically significant under either explanation. The structural-break test for credit growth provides independent evidence consistent with temporal instability (§4.9.7, $p = 0.0066$) and is presented as DGP evidence with reasonable confidence; the term spread's Chow test falls short of significance ($p = 0.379$), so its instability is directionally consistent with DGP non-stationarity but carries less evidential weight, with model variance under the thin early-window sample not excluded. The panel-logistic sign reversal, by contrast, reflects non-monotonic distributed-lag structure and should not be read as evidence of a historical regime break.

Divergence increases across lags: similar shape but different magnitude at lag one, curves separating across the full range at lag two, and at lag three — the highest-ranked EBM feature (importance 1.717) — the pre-GFC model's monotonically increasing risk relationship reversed by the GFC-inclusive model at high spread values. Total credit shows near-zero pre-GFC contributions against increasingly negative GFC-inclusive contributions at high credit values across all three lags. These mappings make one mechanism behind the prospective AUROC collapse (§4.5) legible rather than inferred: pre-GFC mappings misapplied to a structurally different GFC environment produce errors directionally consistent with the observed below-random AUROC, whether the cause is DGP shift or thin-sample estimation — the interpretability contribution, independent of that question.

4.7.4 Hyperparameter Robustness

The baseline EBM configuration (5,000 rounds, 5 interaction terms, learning rate 0.01) was checked against five alternatives varying interaction terms (3, 5, 8) and learning rate (0.005, 0.01, 0.05) on the same specification; AUROC was essentially flat across all five (range 0.0071), and the baseline's own bootstrap 95% CI [0.265, 0.827] contained every alternative's AUPRC point estimate, indicating the baseline was not favourably cherry-picked. Full per-configuration results, including Brier scores, are reported in Appendix B.2.

4.8 Calibration Analysis

Figure 4.8 presents the calibration reliability diagrams for all three Stage 2 models.

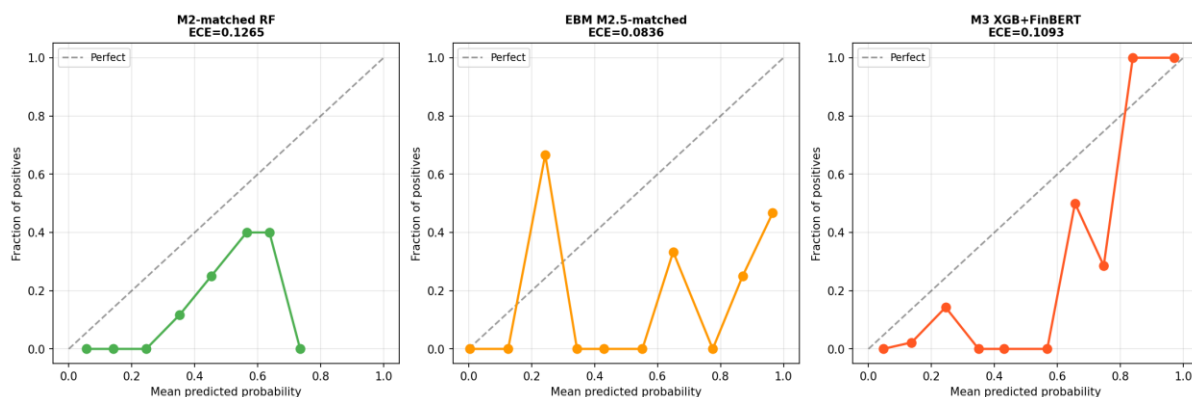


Figure 4.8. Calibration reliability diagrams for Stage 2 models. Left: M2-matched RF ($ECE = 0.127$). Centre: M2.5-matched ($ECE = 0.084$). Right: M3 XGB+FinBERT ($ECE = 0.109$). Dashed diagonal = perfect calibration. Points represent probability bin midpoints vs observed crisis fraction. The noisy appearance reflects 26 positive country-year labels rather than systematic model failure.

Expected Calibration Error for the three 2007 cross-sectional models is 0.127 (M2-matched), 0.084 (M2.5-matched), and 0.109 (M3); Brier Scores are 0.060, 0.067, and 0.061 respectively. EBM achieves the lowest ECE without post-hoc calibration, unlike M3, which applies `CalibratedClassifierCV` within each fold — so this is not a like-for-like comparison. Isotonic calibration was applied to M3 as a fixed component of its pipeline (`CalibratedClassifierCV`, `cv = 2`), reflecting the well-documented miscalibration of gradient-boosted tree probabilities (Zadrozny and Elkan, 2002); it was specified a priori rather than tuned, and was deliberately not applied to RF or EBM so that their native probability calibration could be reported directly — the EBM's native calibration being a property of interest in its own right. Given 26 positive country-year labels and a single episode, neither result is informative about prospective calibration reliability; the parity-fix rationale and the ECE-versus-Brier decomposition are detailed in Appendix B.3.

4.9 Auxiliary Statistical Tests

4.9.1 Multiple-Testing Correction for Auxiliary Statistical Tests

§4.9.2 and §4.9.3 report 84 country-lag Granger tests in total (42 credit growth, 42 term spread), of which 16 are significant at the uncorrected 5 per cent level — above the ~ 4.2 expected by chance (84×0.05), but without correction it is not possible to say which are genuine; citing specific results — such as the five `term_spread_lag3` countries motivating §4.7 — as independently confirmed is not statistically supportable.

Both corrections, computed from NB07B's saved p-values for all 84 tests, leave exactly one survivor: the United States, credit growth, lag 3 ($p = 0.000015$), which clears the Bonferroni threshold ($\alpha/m = 0.05/84 \approx 0.000595$) by a factor of 40 and is the sole Benjamini–Hochberg survivor at $q = 0.05$ (rank $k = 1$, $0.000015 \leq 0.000595$). No other test — including all five term_spread_lag3 country results (§4.9.3) — survives family-wise or false-discovery control. The Bonferroni and Benjamini–Hochberg mechanics are given in Appendix B.3.

Rank k	Country	Lag	Series	p-value	BH threshold ($k/84$) $\times 0.05$
1	USA	3	Credit growth	0.000015	0.000595
2	GBR	3	Term spread	0.001554	0.001190
3	NLD	2	Term spread	0.002601	0.001786
4	NLD	3	Term spread	0.002775	0.002381
5	ITA	3	Term spread	0.009162	0.002976
6	BEL	1	Credit growth	0.009279	0.003571
7	BEL	2	Credit growth	0.010849	0.004167
8	FRA	2	Credit growth	0.012206	0.004762
9	JPN	3	Term spread	0.015128	0.005357
10	DEU	2	Term spread	0.017199	0.005952
11	FRA	3	Credit growth	0.021801	0.006548
12	BEL	2	Term spread	0.023899	0.007143
13	DNK	2	Credit growth	0.024170	0.007738
14	DEU	3	Term spread	0.030462	0.008333
15	BEL	3	Credit growth	0.040921	0.008929
16	ESP	1	Credit growth	0.049143	0.009524

Table 4.9a. Benjamini–Hochberg step-up ladder applied to all 16 nominally significant results in the 84-test Granger causality family (8 credit growth + 8 term spread), computed directly from NB07B's saved output. Only rank 1 (United States, credit growth, lag 3) satisfies its own critical value; the step-up procedure therefore rejects the null for rank 1 only. Bold row indicates the single BH survivor.

Two conclusions follow: this confirms §4.9.3's framing of the term_spread_lag3 concentration as consistent with, not proof of, EBM's importance ranking; and the US credit-growth result at lag 3 is the single most robust finding in the battery, surviving both

corrections by a wide margin — a genuine, not multiple-testing-driven, association (discussed further in §5.1 and §5.5).

This asymmetry (one robust, fifteen not) is itself informative: it shows the battery can produce corrected-significant results when the signal is strong enough, which strengthens rather than weakens the credibility of the null result for the other fifteen.

4.9.2 Granger Causality: Credit Growth

Granger causality tests for credit growth on the extended Stage 1 panel (1989–2020) cover 14 of 18 countries (four excluded: constant crisis indicator makes the F-test uncomputable). Of 42 country-lag combinations, 8 reach nominal significance at the uncorrected 5 per cent threshold — above the ~2.1 expected by chance, but which combinations are genuinely significant rather than false positives cannot be determined without the family-wise correction applied across the full 84-test battery in §4.9.1, which governs interpretation of all individual results below.

Individual combinations are not listed, to avoid selective citation of results that do not survive family-wise control (§4.9.1), except United States lag 3 ($p = 0.000015$), the only result in the 84-test battery surviving both corrections (§4.9.1, §5.5); all other nominally significant credit-growth results are descriptive counts only.

4.9.3 Granger Causality: Term Spread

Granger causality analysis for term spread covers the same 14 countries, excluded for the same reason. Across all three lags, 8 of 42 combinations reach nominal significance at the uncorrected 5 per cent threshold, above the ~2.1 expected by chance; the same correction caveat noted in §4.9.2 applies.

None of the eight nominally significant term spread results — including the five at lag three — survive Bonferroni or Benjamini–Hochberg correction (§4.9.1). The lag-three concentration is a descriptive pattern consistent with EBM's native importance ranking (`term_spread_lag3_dm`, importance 1.717), but reflects shared sample-specific correlation within the GFC-dominated panel, not statistical confirmation of a robust relationship.

The shape function instability in §4.7.3 remains the stronger, more direct evidence of regime non-stationarity at this lag than any Granger pattern here.

4.9.4 Granger Causality: Sentiment

Granger causality tests for FinBERT sentiment on the Stage 2 panel (2003–2020) span six specifications over 26 country-lag pairs each, with significant counts (1/26 to 5/26) at chance level. A Benjamini–Hochberg correction ($q = 0.05$) on NB07B's full 156-test family confirms the null: of 17 nominally significant results (against 7.8 expected), none satisfies its own BH critical value and zero survive Bonferroni, consistent with the non-significant panel logistic increment in §4.9.6.

4.9.5 Panel Logistic: Term Spread

Panel logistic estimation using term spread at all three lag horizons produces a pseudo- R^2 of 0.135 and a statistically significant likelihood ratio ($p < 0.001$), with coefficients reversing sign across lags: `term_spread_lag1_dm` negative and significant ($\beta = -0.646$, $p = 0.004$), `term_spread_lag3_dm` positive and significant ($\beta = +0.630$, $p = 0.003$). This provides parametric evidence that term spread carries different associations across lag horizons, consistent with the temporally non-monotonic pattern in EBM's shape functions: the association between term spread and crisis probability changes sign across the lag horizon, so any model treating it as monotonic across horizons is mis-specified.

4.9.6 Panel Logistic: Sentiment

Panel logistic on Stage 2 data gives a pseudo- R^2 of 0.119 (macro-only) rising to 0.153 with sentiment (increment 0.034), with the `P_neg_lag1` coefficient negative and non-significant ($p = 0.103$). The increment is directional only, consistent with the Granger non-significance in §4.9.4, and a topic-disaggregated extension finds zero incremental AUROC and AUPRC for EBM.

4.9.7 Structural Break Test: Chow-Type Likelihood Ratio

As a formal complement to the shape function evidence (§4.7.3), a Chow-type likelihood ratio test evaluates whether each predictor's relationship to crisis onset differs before and after the 2007 break, using the panel logistic specifications of §4.9.5. For credit growth, splitting at 2007 yields $LR = 14.242$ ($df = 4$, $p = 0.0066$), a significant structural break: the lag-1 rolling-demeaned total loan growth coefficient reverses sign and grows substantially, from -7.17 pre-break to $+28.59$ post-break, and the three-year rolling mean similarly reverses, from $+21.16$ to -4.40 — large, genuine reversals rather than near-zero-baseline artefacts.

For term spread, the same test yields $LR = 4.209$ ($df = 4$, $p = 0.379$): the null of stability is not rejected. Coefficients still shift directionally — lag one from -0.81 to -0.50 , lag two from -0.07 to -0.58 , lag three from $+0.91$ to $+0.72$ — consistent with the shape-function instability in §4.7.3, but the test is underpowered with only 19 crisis events, too few post-break positives at the 2007 split to reach significance. A supplementary 2008 split is discarded: zero post-break positives produce a perfect-separation artefact (nominal $p = 0.0003$), not a genuine result.

Taken together, the credit-growth result provides statistical evidence of a coefficient break at the pre-selected 2007 boundary — subject to sparse-event and panel-dependence limits and to the theoretically rather than data-driven break date — extending the shape-function evidence beyond visual inspection; the term-spread result remains suggestive only.

4.9.8 ML Discrimination vs Statistical Tests

The divergence between ML discrimination and this sparse, country-specific auxiliary-test significance is itself a key finding: the models identify directional AUPRC patterns where

Granger, panel-logistic and attribution methods find no stable relationship — the distinction between in-sample pattern recognition and causal inference, and the reason high expanding-window metrics coexist with prospective failure. The one robust exception is US credit growth at lag 3 (§4.9.1), discussed in §5.5.

4.10 EBM Ensemble Extension

4.10.1 Design and Primary Results

Following Holopainen and Sarlin (2017), an EBM-based ensemble is constructed as $E1-EBM = 0.5 \cdot P(M2\text{-matched}) + 0.5 \cdot P(M2.5\text{-matched})$ using equal weights throughout. Figure 4.9 presents the ROC and PR curves for all ensemble variants.

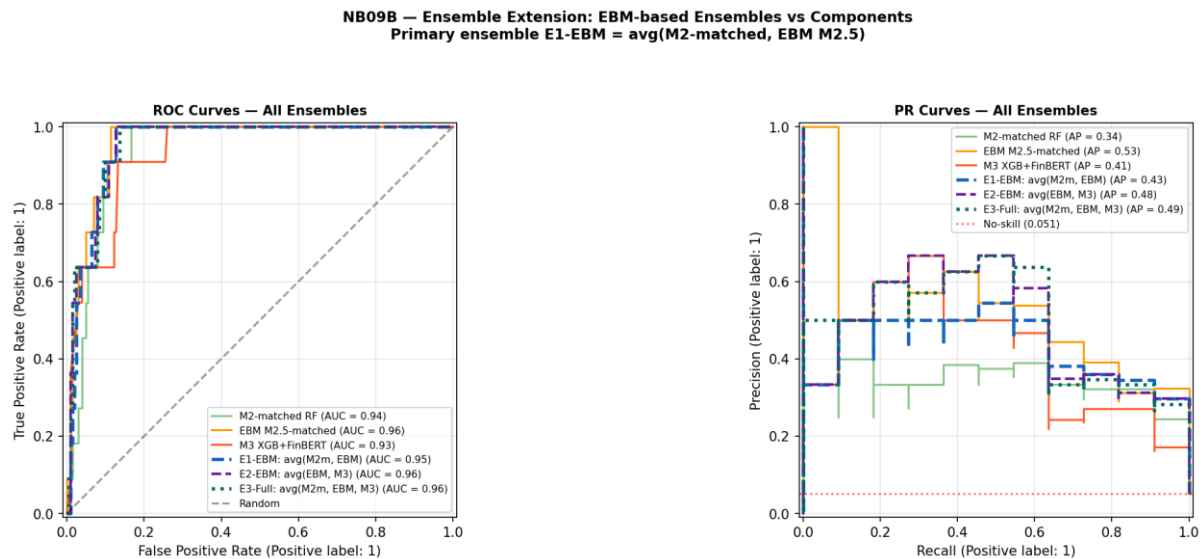


Figure 4.9. Ensemble extension ROC and PR curves (Stage 2 inner-merged scored sample, 11 scored positive labels, base rate 5.1%). Green solid = M2-matched RF; orange solid = M2.5-matched; red solid = M3 XGB+FinBERT; blue dashed = E1-EBM avg(M2m, EBM); purple dashed = E2-EBM avg(EBM, M3); teal dotted = E3-Full avg(M2m, EBM, M3). Red dotted = no-skill baseline. PR curves show that EBM alone (orange) leads on AUPRC; ensemble averaging with lower-AUPRC components reduces the combined AUPRC below EBM's individual level.

Table 4.6 presents the primary comparison on the inner-merged scored sample of fold-generated predictions (215 observations, 11 scored positives, base rate 5.12 per cent) — the same scored sample underlying Table 4.2, so all model rows in the two tables are directly comparable; Table 4.6 adds the ensemble configurations, with confidence intervals from a separate bootstrap run.

Table 4.6. EBM ensemble results — 2007 GFC Case Study (inner-merged scored sample, 11 scored positive labels)

Model	AUROC	95% CI	AUPRC	Brier
M2-matched RF	0.939	[0.894, 0.971]	0.343	0.060
M2.5-matched EBM	0.960	[0.924, 0.986]	0.532	0.067
M3 XGB+FinBERT	0.932	[0.869, 0.979]	0.411	0.061
E1-EBM: avg(M2-matched, EBM)	0.955	[0.918, 0.982]	0.433	0.056
E1-Orig avg(M2m, M3) [M4 ref]	0.943	[0.895, 0.979]	0.440	0.057

Table 4.6. *E1-Orig avg(M2m, M3) is the pre-EBM ensemble retained for reference; it does not correspond to M4 in Table 4.2, which reports E1-EBM avg(EBM, M2-matched). Subject to the evaluation constraints of Section 4.1. E1-EBM improves AUROC (+0.012) and Brier (−0.001) over E1-Orig but is marginally lower on AUPRC (−0.007), reflecting the structural drag of averaging EBM (AUPRC 0.532) with M2-matched (AUPRC 0.343).*

4.10.2 Disagreement Analysis

Figure 4.10 presents the model disagreement analysis for all three model pairs.

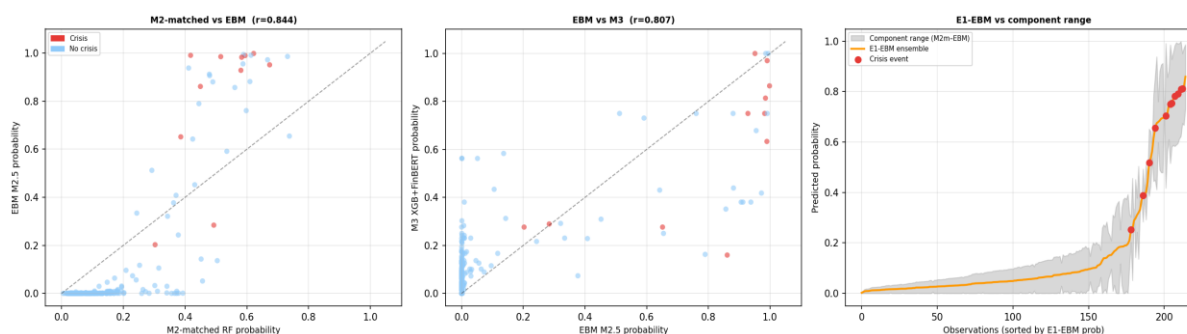


Figure 4.10. Model disagreement analysis (Stage 2). Left: M2-matched RF vs M2.5 (EBM) probabilities (Pearson $r = 0.844$); crisis observations in red. Centre: M2.5 (EBM) vs M3 XGB+FinBERT ($r = 0.807$). Right: E1-EBM ensemble probability vs M2-matched/EBM component range (grey shading); red dots = positive observations sorted by E1-EBM probability.

Mean absolute disagreement between M2-matched and EBM is 0.348 at crisis versus 0.125 at non-crisis observations (ratio 2.8:1), and between EBM and M3 it is 0.209 versus 0.113 (ratio 1.8:1) — substantially higher at crisis observations in both pairs, consistent with the

Holopainen and Sarlin (2017) motivation that ensemble averaging is most valuable where individual uncertainty is highest. Pearson correlations of 0.844 (M2-matched vs EBM) and 0.807 (EBM vs M3) indicate shared signal alongside meaningful complementarity.

4.10.3 2007 Cross-Sectional Ranking

This crisis-type analysis is explicitly exploratory and post-hoc: the Fundamental/Contagion/No-crisis classifications are applied after model training and were not pre-specified as a formal test, so the findings are descriptive and hypothesis-generating rather than validated capabilities.

Figure 4.11 presents the E1-EBM 2007 country ranking by crisis type.

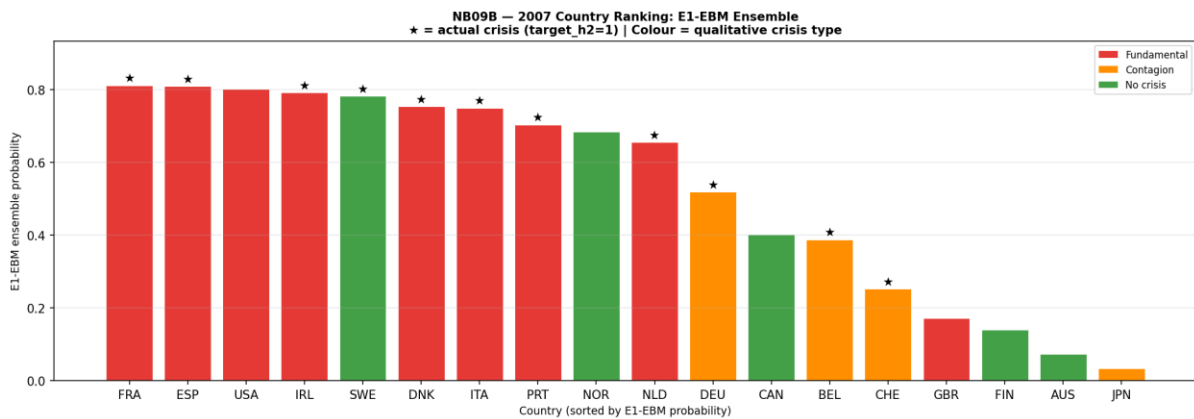


Figure 4.11. 2007 country probability ranking under E1-EBM ensemble, sorted descending. Red = Fundamental crisis; orange = Contagion; green = No crisis. Stars (★) indicate target_h2 = 1 (crisis onset within two years of 2007). Mean E1-EBM by type: Fundamental 0.695 (n=9), Contagion 0.395 (n=5), No crisis 0.325 (n=4). Sweden reclassified as Contagion following JST Macrohistory Database R6 confirmation of banking crisis onset 2008 (target_h2 = 1 in 2007); crisis mechanism was international contagion rather than domestic credit accumulation. The ordering Fundamental > Contagion > No crisis reflects the model's training on domestic macro credit-cycle precursors.

Two country-level results warrant qualification. Sweden (E1-EBM 0.783, rank 5, Contagion ★) is a true positive: JST R6 records banking crisis onset in 2008 with target_h2 = 1 in 2007; the high model probability reflects macro precursor signal despite the crisis mechanism being international contagion rather than domestic credit accumulation. The United Kingdom (E1-EBM 0.171, rank 15, No crisis) receives a low probability and carries no target_h2 flag in 2007 under the JST chronology boundary used in this study; this result reflects the scope conditions of the labelling convention rather than a statistical false negative and is discussed further in Chapter 5.

4.11 Hypothesis Assessment

Table 4.7. Hypothesis assessment under the augmented two-stage pipeline with EBM extension

Hypothesis	Stated prediction	Key evidence	Verdict
H1: Discrimination	M3 achieves higher AUROC/AUPRC than macro-only benchmark	2007 GFC cross-sectional country ranking: M3 AUPRC 0.411 vs M2-matched 0.343 — directional only, CI $[-0.130, +0.312]$ spans zero. matched pairwise decomposition (§4.2.1): holding algorithm fixed at EBM, sentiment reduces AUPRC by -0.078 (bootstrap 95% CI $[-0.292, +0.187]$, spans zero); holding features fixed at 61, EBM outperforms XGBoost by $+0.043$ (no independent CI computed). Both effects derive from the same single cross-sectional fold; the decomposition is non-orthogonal (Table 4.1). Directional observations, not confirmed differences.	Not confirmed. Directional only (CI spans zero; single-fold, non-orthogonal decomposition); corroborated independently by §4.9's auxiliary tests (0 of 156 Granger tests, panel logistic non-significant).
H2: Feature relevance	At least one FinBERT feature ranks top-5 by SHAP	5 sentiment features in top 20 ² ; SHAP share 26.2%. EBM native importance corroborates macro dominance independently. Agreement between methods reflects shared sample-specific correlation structure, not causal evidence. All statistical tests non-significant.	Not supported in strict formulation; weakly and conditionally observed via convergent SHAP/EBM macro dominance within the GFC sample.
H3: Interpretability trade-off	Quantifiable accuracy-interpretability trade-off across M1–M2–M3	M1→M2: AUPRC $+0.085$ at H2 with loss of log-odds transparency. M2→M2.5: within expanding-window evaluation, interpretable EBM appears to achieve higher point estimates than black-box RF. EBM collapses more severely under prospective validation (AUROC 0.333 vs RF 0.481), demonstrating that interpretability does not confer prospective robustness and may increase vulnerability to regime-specific overfitting.	Partially supported (non-monotonic); the trade-off is not binary within the non-linear model class; the flexibility extension is exploratory, not confirmatory (see discussion below).

Table 4.7. Hypothesis assessment under the augmented two-stage pipeline (Notebooks 01-NB09B). Stage 1 AUPRC at H2 is the primary metric for multi-fold OOS evaluation. Stage 2

² This ranking is derived directly from the trained M3 model, which uses the actual 61-feature set (41 macro + 20 sentiment) described in §3.3, not the intended 14-feature sentiment specification. The 20-vs-14 discrepancy is therefore a documentation-labelling issue only and does not affect this ranking or the five-of-20 count: both are properties of the model as estimated. Expressed against the intended 14 sentiment features, five of top 20 would read as a larger 35.7% share rather than 25.0%; this framing was not used, as it would overstate sentiment's relative prominence relative to the feature set the model was actually trained on. Either way, the H2 verdict of non-support is unaffected, since no sentiment feature ranks in the top five by SHAP or EBM native importance.

verdicts reflect country ranking within the 2007 GFC cross-sectional case study only — they are not OOS predictive results and should not be read as confirmation of prospective model performance.

The three hypotheses together trace a coherent argument about the structural limits of ML-based crisis prediction: H1 is directionally supported, but case-study performance differences reflect algorithm non-linearity at least as much as sentiment signal; H2 fails its strict formulation, since SHAP/EBM agreement reflects shared sample correlation, not causal validation; H3 is partially supported, but not in the originally anticipated monotonic form: interpretability does not confer prospective robustness and may increase vulnerability under regime change.

The refinement of H3 — that trade-off severity varies within the non-linear class, with EBM's flexibility linked to sharper prospective collapse under structural break (§4.5.1, §4.7.3, §4.9.7) — was identified during analysis rather than pre-specified, and is reported as a data-driven refinement, not a confirmatory test.

Chapter 5: Discussion of the Results

5.1 Overview of Findings

This chapter synthesises Chapter 4's results, situates them in the EWS literature, and draws out implications for research design, practical application, and the theoretical limits of ML-based crisis prediction. Four principal findings, together with two supplementary results and one auxiliary statistical finding, define the augmented analysis's contribution. The EBM interpretability benchmark is the most theoretically significant, testing whether native interpretability confers any robustness advantage under structural non-stationarity.

Extending macro training history to 1989–2020 does not resolve the EWS generalisation problem. M2 achieves an AUROC of only 0.481 in strict temporal validation (§4.5.1), marginally below random, which is consistent with the interpretation that extending historical training data alone does not overcome structural regime discontinuity.

A second finding is that sentiment augmentation shows no reliable signal: M3's directional AUPRC gain over M2-matched has a confidence interval spanning zero, no auxiliary

sentiment test survives BH correction, and once algorithm class is held fixed at EBM, sentiment reduces rather than improves discrimination (§4.2, §4.2.1, §4.9.4, §4.9.6).

The EBM interpretability benchmark provides the third finding: M2.5 exceeds M2 under the Stage 1 expanding-window design, while M2.5-matched exceeds M3 within the separate 2007 cross-sectional case study, in both cases while remaining natively interpretable. Its collapse in prospective Split B testing (AUROC 0.333 against RF's 0.481) is diagnostic rather than conclusive, given only six pre-GFC training positives; the shape function analysis (§4.7, §5.2) provides more direct evidence of regime-specific instability, not relying on performance differences from a tiny positive sample.

As a supplementary result, the ensemble extension (E1-EBM) achieves the lowest Brier Score of any case-study configuration (0.056) at AUROC 0.955, within the same 2007 cross-sectional scope as all other case-study results.

A second, exploratory supplementary result is that E1-EBM assigns systematically higher probabilities to fundamental credit-boom crises than to contagion-driven or no-crisis countries. This pattern is post-hoc rather than pre-registered, but it motivates the case study in §5.4.

One auxiliary finding stands out: of 84 Granger tests across credit growth and term spread, only US credit growth at lag 3 survives family-wise correction ($p = 0.000015$, §4.9.1) — the sole statistically robust predictor-outcome relationship in the battery.

The fourth principal finding synthesises the rest: taken together, the findings are most consistent with structural instability in the predictor–outcome mapping across periods — within which the US credit growth channel provides the most statistically durable auxiliary evidence — although estimation instability arising from sparse crisis samples cannot be excluded. §5.2 develops the theoretical implications.

5.2 The EBM Interpretability Benchmark: What Native Transparency Reveals

The introduction of M2.5 (EBM) represents the most theoretically significant extension of the original study. Its contribution lies not in the performance metrics, which are subject to the same evaluation constraints as all other models, but in what its shape functions reveal about the structure of the learning problem.

Because EBM's shape functions are the model itself rather than a post-hoc approximation, any instability observed in them cannot be attributed to approximation error; that confound is eliminated by construction. What remains is a property of the fitted model itself, consistent with instability in the data-generating process, though regime-specific variance under thin sub-period samples cannot be fully excluded (§4.7.3). The structural break test (§4.9.7) provides the more direct evidence for the DGP interpretation; the panel logistic reversal

(§4.9.5) reflects non-monotonic lag structure and is consistent with, but does not by itself establish, that interpretation. Figure 4.7 shows this directly: the learned feature-risk mapping is training-window-dependent at every lag examined, with pre-GFC and GFC-inclusive models producing qualitatively distinct, crossing shapes. For term spread, the same feature value carries reduced crisis risk under the early GFC-run-up window (2003–2006) and increased risk under the GFC-inclusive window, a training-window sign inversion that no static, pooled model can represent. The instability is a direct, readable property of the fitted model.

The pattern places crisis prediction within the broader class of non-stationary ML problems under distribution shift, strengthened without being fully confirmed by the corroborating evidence above and the regime-conditional prototype (Appendix A). Subject to that qualification, static pooled EWS designs face a structural challenge that more data, better algorithms, or improved interpretability cannot resolve alone — the architecture itself may be misaligned with a non-stationary environment.

A second, complementary strand of evidence for regime non-stationarity comes from the regime-conditional prototype (Appendix A), and is important enough to state in the main analysis rather than leave to the appendix. When the EBM is refitted on the pre-GFC sample alone (1989–2002), the entire feature hierarchy changes: the three lags of the term spread, which dominate the importance ranking in the pooled model fitted on the full panel (lag 3: 0.791), all but disappear from the pre-GFC ranking, where the term spread first appears only at rank 7 (0.147). In their place the leading predictors become inflation and the government debt-to-GDP ratio (CPI lag 2: 0.352; debt-to-GDP lag 3: 0.310). This shift is economically coherent: the early-1990s Nordic crises and Japan 1997 followed episodes of high inflation, rapid credit growth and fiscal deterioration (Reinhart and Rogoff, 2009), whereas the GFC was preceded by term-spread compression under low inflation (Borio and Lowe, 2002) — so the pooled model’s apparent reliance on the term spread is largely an inheritance from the GFC observations that supply most of its positive cases. The finding matters because it is not a statement about shape but about which features the model relies on at all: a single pooled classifier trained across regimes may learn a feature hierarchy that is unrepresentative of one or more constituent regimes, which is precisely the failure mode that the additional data and flexibility tested here did not overcome.

Two further results from the same prototype belong in the main analysis. First, regime-conditioning does not by itself rescue prospective performance: fitted on the manually defined pre-GFC and GFC regimes, every configuration (Pooled, R1 and R2) still fails the 0.75 AUROC benchmark and falls below the no-skill precision–recall baseline (Appendix A.3), so the prototype does not establish that regime conditioning improves prediction; it demonstrates the feasibility of the approach and shows that reliable evaluation is prevented by the scarcity of independent crisis observations within each regime. Second, the instability is asymmetric rather than merely large: the term spread carries almost no signal in the pre-GFC regime yet an extreme, threshold-driven effect in the GFC regime (Appendix A.5) — an asymmetry no single pooled shape function can represent.

SHAP-EBM convergence on `term_spread_lag3_dm` (§4.6.3) is methodological triangulation within the 2007 cross-section, not independent confirmation; the nominally consistent Granger pattern at that lag (5 of 14 countries, §4.9.3) does not survive correction (§4.9.1) and is not cited as corroboration. The shape function instability (§4.7.3) remains the stronger, more direct evidence, depending on neither.

The panel logistic sign reversal, with the lag-one term spread coefficient at $\beta = -0.646$ ($p = 0.004$) against a lag-three value of $\beta = +0.630$ ($p = 0.003$), provides independent parametric evidence that term spread has different associations across lag horizons, which a single monotonic contemporaneous effect would not capture; it is a distributed-lag sign difference rather than direct evidence of change across historical regimes, and cannot be captured by any model that treats it as fixed.

For EWS design, this means that a model trained on one crisis regime may systematically misread signals in a structurally different regime when the relevant feature–risk mapping has changed.

5.3 The Ensemble as a Response to Model Uncertainty

The ensemble extensions operationalise the Holopainen and Sarlin (2017) insight that combining architectures beats single-model selection under small-sample, crisis-concentrated evaluation. Equal weighting is used throughout, since performance-weighted schemes are circular when weights derive from the same out-of-sample pool used for evaluation.

The disagreement analysis (§4.10.2) provides the substantive justification: correlation between M2-matched and EBM is high ($r = 0.844$) but concentrated in low-risk agreement, while disagreement is 2.8 times higher at crisis than non-crisis observations, so the ensemble's moderating effect is strongest precisely where single-model overconfidence is most costly.

E1-EBM's Brier Score of 0.056, the best of any 2007 GFC case-study configuration, comes at the cost of lower AUPRC than EBM standalone (0.433 vs 0.532): the ensemble suits policy-threshold calibration, while EBM alone suits within-episode ranking. The trade-off itself reflects regime instability, since averaging moderates the extreme probabilities EBM assigns under aggressive regime-specific fitting, trading discrimination for calibration. That no single configuration dominates every criterion motivates the country-level analysis in §5.4, where E1-EBM's crisis-type signal is examined directly.

5.4 Crisis Heterogeneity and Model Scope

A structural limitation across the EWS literature, including this analysis, is that crisis targets from banking-crisis chronologies are outcome-based rather than mechanism-based. The Laeven and Valencia (2020) chronology records when a systemic crisis occurred but does not distinguish domestic credit-boom, contagion, and exogenous-shock crises. Training on this undifferentiated label implicitly assumes the same features are relevant across all crisis types — an assumption this chapter's evidence consistently refutes.

Four crisis mechanisms are relevant: (1) fundamental crises driven by domestic credit booms and asset price accumulation (Nordic cluster, Japan 1990s, US/Ireland/Spain pre-GFC), the type EWS macro features are designed to detect; (2) contagion crises propagated through financial linkages without domestic imbalance (Germany, Belgium, Switzerland 2007–08); (3) speculative attacks against fixed exchange rates (ERM 1992; Obstfeld, 1996); and (4) exogenous real shocks such as COVID-19, appropriately excluded from the crisis chronology. Training on an undifferentiated target that conflates all four introduces noise into the learning problem.

The 2007 cross-sectional ranking under E1-EBM (§4.10.3: fundamental 0.695 vs contagion 0.395 vs no-crisis 0.325) provides qualified, exploratory evidence of mechanism-specific discrimination — post-hoc, not pre-specified — coherent with the model's design: macro features capturing credit, rate, and housing dynamics carry more signal for domestically generated imbalances than externally transmitted stress. The corresponding M4 finding in the original analysis (fundamental mean 0.599, contagion 0.281) shows the same directional ordering.

The country-level labels that follow — fundamental crisis, contagion, and no crisis — refer to qualitative mechanism classification and should be distinguished from the binary JST target_h2 evaluation label. Two country-level results warrant explicit discussion: Sweden (a true positive whose elevated precursor signals overlap with contagion exposure) and the United Kingdom (a qualitative scope failure, target_h2 = 0 in 2007, examined in Section 5.4.1). Both illustrate the limits of the macro precursor approach.

5.4.1 Feature-Level Evidence: Spain, Sweden, and the United Kingdom

The 2007 cross-sectional ranking provides the basis for a structured feature-level case study across three countries spanning the range of model outcomes: Spain (correct hit; E1-EBM 0.810), Sweden (true positive; E1-EBM 0.783), and the United Kingdom (qualitative scope failure, target_h2 = 0; E1-EBM 0.171). Examining the macro feature values the model observed for each country in 2004–2006 — the lagged window feeding the 2007 prediction — illustrates both the EBM's discriminatory logic and the scope conditions under which it fails.

Table 5.1. Case study feature evidence — Spain, Sweden, and the United Kingdom

Feature	Spain	Sweden	United Kingdom
Role	Correct hit	True positive†	Qualitative scope failure
E1-EBM probability	0.810	0.783	0.171
Credit growth, raw (% p.a.)	19.8	7.4	9.3
House price growth (% p.a.)	17.6	11.1	13.3
Term spread lag 1 (dem., pp)	1.280	1.069	0.538
Short rate lag 1 (dem., pp)	-6.77	-5.60	-3.57

Averages over 2004–2006, lag 1. Demeaned = deviation from expanding country-specific historical mean.
† Sweden: JST R6 records crisis onset 2008 (target_h2 = 1 in 2007); mechanism is contagion, not domestic credit boom.
Source: JST Macrohistory Database R6; NB09B ensemble output.

Figure 5.1a presents credit and housing conditions averaged over 2004–2006. Spain's credit growth (19.8% p.a.) and house price growth (17.6%) are consistent with the classic pre-crisis domestic boom pattern (Schularick and Taylor, 2012); EBM shape functions assign increasing crisis probability at elevated credit growth, and Spain sits well into the high-risk segment. Sweden's credit growth (7.4%) and house price growth (11.1%) are moderate; the model's high probability for Sweden (0.783) reflects overlap on yield-curve and monetary accommodation signals, which are not mechanism-specific and do not discriminate domestic boom from contagion exposure.

The United Kingdom is more precisely a scope condition than a model error. UK 2007–08 stress originated on the liability side, in short-term wholesale funding of mortgage books, rather than in asset-side imbalance accumulation. The model's inputs (credit growth 9.3%, house price growth 13.3%) correctly receive low crisis probability; the crisis mechanism (funding-channel fragility) lies outside the feature set's scope. This is a scope-related false negative rather than an unexplained failure: the model does not flag the case, but the omission is attributable to funding-channel fragility lying outside the macro precursor scope.

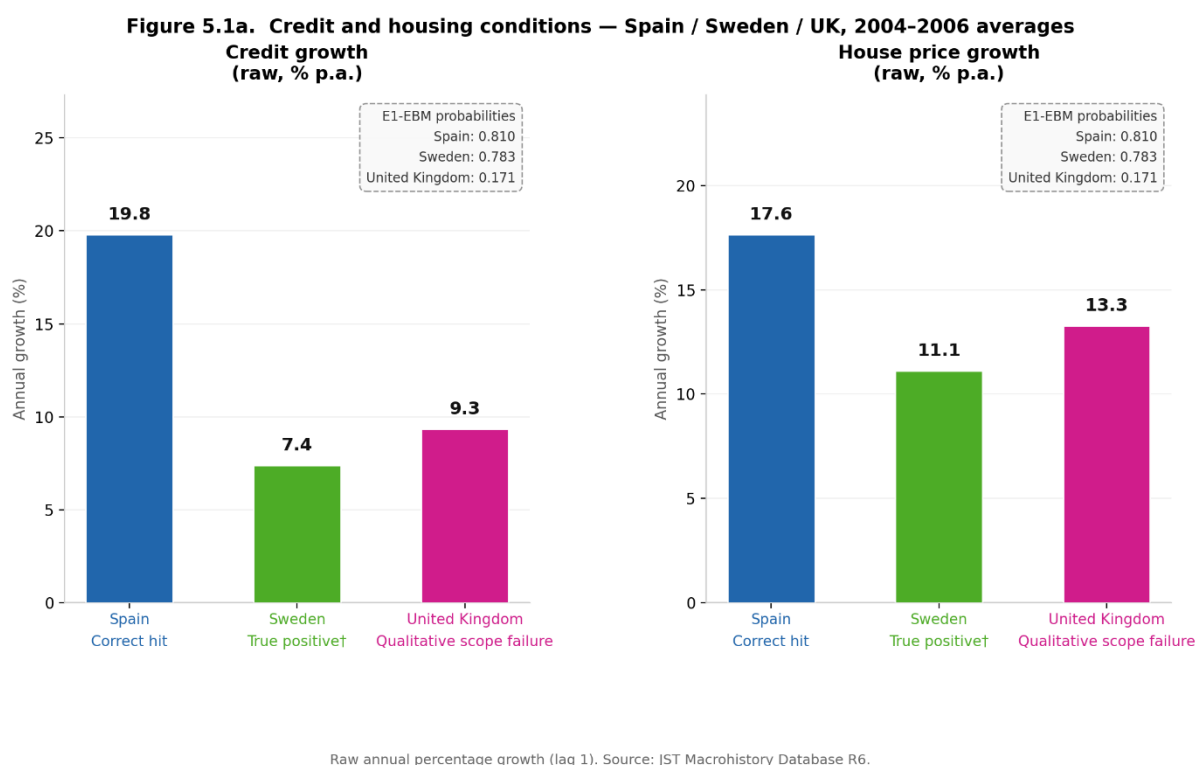
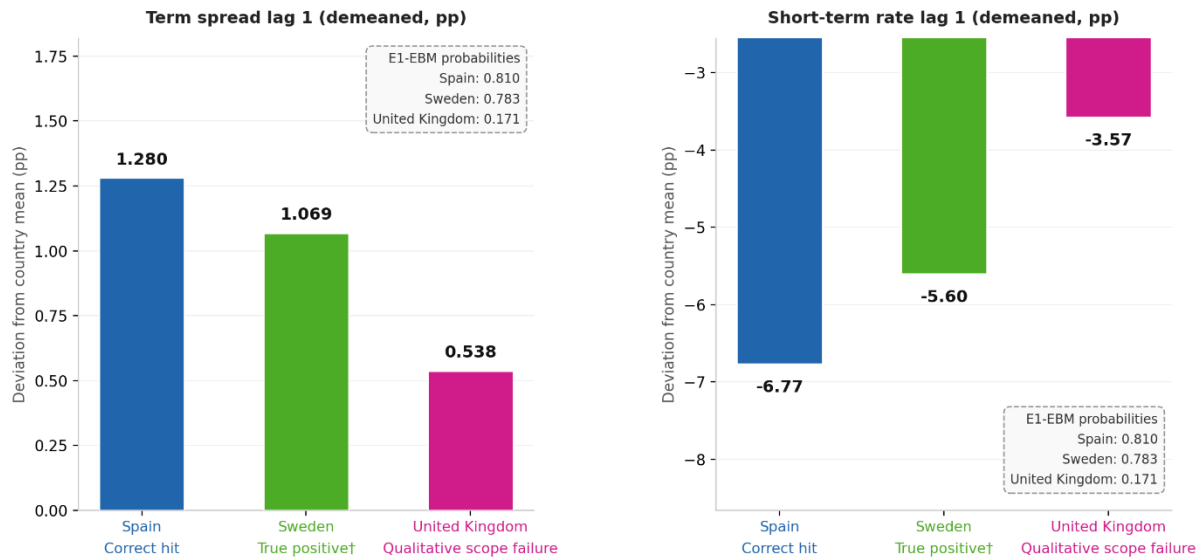


Figure 5.1b presents the monetary and yield-curve conditions, expressed as deviations from each country's expanding historical mean at lag 1. All three countries display term spreads above their historical averages (Spain 1.280 pp, Sweden 1.069 pp, United Kingdom 0.538 pp) and short-term rates substantially below their historical averages (Spain -6.77 pp, Sweden -5.60 pp, United Kingdom -3.57 pp). The directional pattern is consistent across all three: highly accommodative monetary conditions relative to each country's history, with steepened yield curves, reflect the shared low-rate environment of the mid-2000s rather than country-specific imbalance. This limits the cross-sectional discriminatory power of monetary and yield-curve features: they raise aggregate crisis probabilities across the sample but do not separate Spain's domestic boom from Sweden's contagion exposure or the UK's funding-channel vulnerability.

**Figure 5.1b. Monetary and yield-curve conditions
Spain / Sweden / United Kingdom — 2004-2006 averages**



Source: JST Macrohistory Database R6. † Sweden: crisis onset 2008 (target_h2 = 1 in 2007); contagion mechanism — true positive.

The contrast makes the model's logic visible without post-hoc attribution: Spain stands apart on credit and house-price growth (two to three times Sweden's and the UK's), while all three converge on monetary conditions, which add little cross-sectional discrimination. The signal concentrates in credit-cycle dynamics and the errors have identifiable mechanistic causes — that legibility is EBM's interpretability contribution.

5.5 Implications for Early-Warning System Design

Five practical implications follow from the augmented analysis, stated with appropriate caution given the evaluation constraints throughout Chapter 4.

Extending training history is necessary but not sufficient: the shape function instability (§5.2) indicates that, within the additional historical sample examined here, regime heterogeneity is a more important constraint than training-window length alone, so an operational EWS needs a regime-classification layer or a portfolio of mechanism-specific models rather than a single general-purpose classifier.

Interpretable non-linear models such as EBM offer a capability black-box models cannot: direct observation of whether learned feature-risk relationships are regime-stable (Figure 4.7), whereas post-hoc SHAP instability may simply reflect approximation error. That transparency value is distinct from prospective performance — Split B shows higher

flexibility can worsen regime-specific overfitting — so EBM should be adopted for interpretability, not as a claim of prospective superiority.

The ensemble design, meanwhile, is most defensible for calibration rather than discrimination: E1-EBM's Brier Score (0.056) is the lowest of any configuration, which matters for policy threshold-setting, though the advantage is observed once, in 2007, and may not generalise.

Fourth, central bank sentiment does not contribute a robust signal (§4.2.1, §4.9.4, §4.9.6): the apparent M2-matched-to-M3 gain reflects an algorithm switch, not a sentiment signal, and is best treated as hypothesis-generating rather than an established predictive channel.

Fifth, US credit growth at a three-year lag is the only auxiliary predictor surviving rigorous multiple-testing correction (§4.9.1), consistent with the credit-boom literature (Schularick and Taylor, 2012). This is the only statistically robust auxiliary justification for including a multi-year credit growth lag; the term spread pattern, despite its prominence in EBM and SHAP attribution, should be treated as a within-episode correlation, not a confirmed signal.

5.6 Practical Recommendations

Four practical recommendations follow from the analysis and discussion above, addressed to practitioners and researchers involved in the design and deployment of financial early-warning systems. Each is deliberately scoped to what the evidence and its mitigations (§5.7) support, and is framed so as not to overreach the single-episode, regime-specific basis on which it rests.

Recommendation 1: Prospective validation across crisis regimes must be a mandatory evaluation step.

Expanding-window metrics substantially overstate prospective predictive power under this kind of cross-sectional evaluation, so strict temporal validation, meaning training exclusively on pre-crisis data and evaluating on a subsequent episode, should be mandatory, not optional; models failing it must not be characterised as predictively useful without explicit disclosure.

Recommendation 2: Interpretable non-linear models should be used as primary diagnostic benchmarks and considered alongside black-box alternatives, rather than presumed to be prospectively superior.

In this study, EBM achieved higher expanding-window point estimates than the RF benchmark while providing natively interpretable shape functions requiring no post-hoc approximation. Shape function stability analysis, meaning whether the learned feature-risk mapping is consistent across training sub-periods, should be a mandatory diagnostic step, and feature selection should be informed by rigorous multiple-testing correction: in this study, US credit growth at a three-year lag is the only feature with a statistically confirmed auxiliary association after family-wise correction across 84 tests, a stronger justification than within-episode ML attribution rankings alone. Supervisory and model risk management frameworks

should treat shape-function stability as a first-order diagnostic criterion, while noting that the EBM degrades furthest in the one informative strict temporal split.

Recommendation 3: Ensemble averaging should be considered as a probability-smoothing output mechanism.

When individual EWS models cannot be definitively ranked on prospective performance, equal-weighted ensemble averaging (weights independent of the out-of-sample pool, avoiding circularity) achieves the lowest Brier Score of any configuration. This is an exploratory probability-accuracy benefit within the 2007 cross-section rather than demonstrated prospective reliability, and it comes at the cost of within-episode discrimination: AUPRC drops relative to the best single model (E1-EBM 0.433 vs EBM standalone 0.532), so the appropriate choice depends on whether the priority is overall probability accuracy or within-episode ranking.

Recommendation 4: Sentiment augmentation requires multi-episode validation before operational deployment.

The directional AUPRC gain observed in the 2007 GFC case study is sample-specific and statistically unconfirmed; validation across multiple structurally distinct crisis episodes is required before it can be treated as generalisable. Topic-specific text features (Ho, Moench and Steeley, 2022) are a more promising avenue than aggregate sentiment polarity, but the completed topic-sentiment extension here finds no measurable incremental gain over the macro baseline, reinforcing the case for multi-episode validation.

5.7 Limitations

Several limitations constrain the interpretation of results throughout this analysis.

The evaluation structure itself is the dominant constraint: the case-study evaluation rests on crisis events in a single year (2007) across 18 countries, 11 positives at a 0.0512 base rate, so AUPRC is materially sensitive to small rank changes and the differences in Table 4.2 should be read as directional rather than stable. The strict temporal validation (§4.5), producing AUROC below random for both M2 and EBM, remains the empirically grounded bound on genuine forward-looking deployment. The same constraint applies reflexively to the Stage 1 interpretability finding: while the EBM–RF gap is robust to cross-country resampling, it is not robust to resampling by year (§4.2), because 93 per cent of scored Stage 1 positives originate in the 2007–08 episode. No result in this study, including its own headline comparison, should be read as established across crisis regimes.

Replicating the case-study design on a second multi-country episode would be the direct test of whether these results are specific to the Global Financial Crisis, and that replication is not available within this panel. The JST Macrohistory Database covers 18 advanced economies,

so the other candidate multi-country episodes of the modern period fall largely outside its country coverage, the 1997 Asian crisis being the clearest example, while the 1992 ERM episode is a speculative-attack mechanism rather than a banking-crisis chronology entry. Conclusions drawn from the 2007 cross-section are accordingly hypothesis-generating about early-warning behaviour under one structural regime, not confirmed findings about early-warning systems in general, and they are reported on that basis throughout.

The crisis label heterogeneity affects the mechanism-discrimination finding in particular: training on an undifferentiated target that conflates fundamental, contagion, and exogenous shock episodes introduces noise, so the 2007 cross-sectional discrimination result is suggestive but cannot be validated without a mechanism-labelled training target — a limitation rather than an oversight.

The EBM interpretability contribution is qualified by the Split B prospective collapse (AUROC 0.333). Interpretable models should be adopted for their transparency value; higher in-sample performance should be treated with the same scepticism applied to all expanding-window metrics.

The class of models evaluated is also bounded. The comparison spans regularised logistic regression, tree ensembles and a gradient-boosted generalised additive model, but excludes recurrent architectures such as long short-term memory networks, which Tölö (2020) applies to crisis prediction and which are designed explicitly to exploit the temporal ordering of the input series. The finding that neither algorithm generalises prospectively (Section 4.5.1) therefore holds for the architectures tested rather than for machine learning as a class, and the omission is acknowledged here as a scope condition on that claim. Two results nevertheless bound what a recurrent model could plausibly recover. The shape function reversals documented in Section 4.7.3 and the significant Chow-type structural break in credit growth at 2007 (Section 4.9.7; LR = 14.242, $p = 0.0066$) indicate that the feature-outcome mapping itself inverts across the sample, rather than that a stable mapping is being missed because its temporal structure goes unmodelled. A sequence model learns dependency structure within a mapping; it does not repair a mapping whose sign reverses. Non-stationarity of this form is a property of the data-generating process rather than a deficiency of architecture, and testing that proposition directly, by running a recurrent model through the same Split B design, is left to future work.

The sentiment corpus is unevenly distributed: BIS speech coverage varies by volume and linguistic register across central banks, so countries with smaller English-language corpora receive less stable FinBERT scores; rolling demeaning partially addresses temporal non-stationarity but does not equalise cross-country corpus quality.

Bootstrap confidence intervals throughout Chapter 4 characterise sampling variability in the pooled predictions only; panel dependence (crisis events clustered within the same episode) means they understate true population-level uncertainty, and are retained as descriptive summaries only. The disagreement analysis (§4.10.2) offers a better-grounded characterisation, not resting on an independence assumption.

These limitations were addressed as far as the design permits. Small-sample inferential fragility is mitigated — though not removed — by three measures in Chapter 4: bootstrap confidence intervals on the headline effects, a family-wise multiple-testing correction across

the full Granger battery (§4.9.1), and the disagreement analysis (§4.10.2), which characterises model behaviour without the independence assumption the confidence intervals require. Country dependence is mitigated by leave-one-country-out validation (§4.5.2), which shows the interpretability gap is robust to dropping any single country, even though it is not robust to dropping the GFC year. Specification risk in the EBM is mitigated by the hyperparameter-sensitivity check (Appendix B.2), whose alternative configurations all fall inside the baseline’s confidence interval. Uneven sentiment-corpus quality is partially mitigated by rolling demeaning and is otherwise disclosed as a bounded limitation rather than corrected. Two constraints, by contrast, are not mitigable within this dataset and are therefore treated as boundaries on what the study can claim: the concentration of scored positives in the single 2007 episode, and the undifferentiated crisis label. These are the reasons the strong findings are stated as regime-specific and the below-random prospective results are treated as the binding evidence — a discipline carried through into the conclusions and recommendations.

5.8 Extensions and Future Research

Three extensions follow directly from the present analysis, ordered by feasibility and proximity to the existing framework.

Robustness Check: Topic-Specific Sentiment Features

Following Ho, Moench and Steeley (2022), who show topic-specific measures produce more granular, stable textual features than aggregate indices, this extension replaces generic FinBERT polarity with topic-disaggregated signals: a keyword taxonomy assigns the 16,622 BIS speeches to three domains (credit conditions, monetary policy, corporate sector; real estate and sovereign excluded for coverage), with FinBERT negative sentiment computed per topic and lagged one to three years, generating nine additional features. The detailed results are produced by notebook NB10C, listed in the Appendix B pipeline inventory.

Topic disaggregation does not recover a signal aggregate sentiment fails to provide: EBM shows no measurable incremental AUROC or AUPRC. Topic-sentiment features exhibit non-stationarity consistent with §4.7.3’s macro findings — monetary-policy sentiment at lag three most unstable, corporate sentiment at lag one most stable. The topic-to-macro importance ratio (0.240) quantifies the macro dominance documented in §4.6. Regime instability appears to be a property of the crisis-generating process spanning both macro and textual features; topic disaggregation does not resolve the generalisation limitation.

Regime-conditional modelling (highest impact; exploratory proof-of-concept implemented in Appendix A)

The shape function instability (§4.7.3) motivates a two-stage, regime-conditional design in place of a single static classifier. Stage 1 trains separate EBMs on disjoint historical sub-

periods (e.g. 1989–1997 Nordic crisis, 1998–2006 pre-GFC Great Moderation, 2007–2012 GFC and aftermath) and clusters their vectorised shape functions to identify candidate regimes from the data rather than imposing them exogenously. The sign reversal already visible in Figure 4.7 is preliminary evidence that such clustering may recover meaningful structure.

Stage 2 retrains a separate EBM per discovered regime, with a master classifier routing new observations by predicted regime membership; this directly tests the dissertation's core thesis, since conditioning prediction on regime should, in principle, outperform any static pooled classifier.

Such a system would be a direct empirical test of the non-stationarity thesis against the Split B baselines (EBM AUROC 0.333, RF 0.481) and ideally the 0.75 benchmark. The exploratory proof-of-concept in Appendix A, using manually defined pre-GFC and GFC-inclusive regimes, demonstrates feasibility while confirming the binding statistical constraints preventing a fully validated system within the current dataset.

A related but architecturally distinct extension conditions on crisis mechanism rather than on time. Where the design above routes observations by discovered temporal regime, a mechanism-conditional system would place a first-stage classifier over crisis type, separating fundamental, contagion and speculative-attack episodes, and route each observation to a specialist early-warning model trained only on episodes of that type. This would test the heterogeneity argument of Section 5.4 at the modelling level rather than the diagnostic level. The obstacle is sample size rather than concept: the 19 crisis onsets in the panel, distributed across the four mechanisms set out in Section 5.4, leave roughly nine fundamental and five contagion positives, too few to fit either stage credibly. The mechanism labels used in Section 4.10.3 are also assigned post hoc by the analyst rather than drawn from an independent chronology, so a wider panel carrying externally sourced mechanism codings would be a precondition for the design.

Network contagion features (most ambitious)

Incorporating bilateral bank exposure data from the BIS consolidated banking statistics, following Cetorelli and Goldberg (2011), would let the EWS framework capture cross-country financial linkages currently invisible to the model, plausibly improving the weak, inconsistent performance seen for contagion-driven cases such as Germany, Belgium, and Switzerland in 2007–2008 (§5.4). This is the most ambitious extension: it faces missing bilateral exposures, temporal-alignment challenges, and comparability issues between residence-based macro indicators and consolidated banking-sector positions, requiring substantial additional data collection and panel integration.

5.9 Summary

The augmented analysis, extended with the EBM interpretability benchmark and ensemble, advances the original study on four fronts. Empirically, the prospective failure of macro-based EWS is not an artefact of insufficient training data. Methodologically, it exposes the evaluation bias inherent in expanding-window validation and shows ensemble averaging improves global calibration (Brier 0.056) even where discrimination gains are statistically indistinguishable. From an interpretability standpoint, EBM's shape functions convert an opaque model failure into a directly observable, regime-specific misspecification.

Theoretically, EBM's more severe prospective collapse relative to Random Forest in Split B is directionally consistent with greater regime-specific overfitting under higher flexibility, though treated as indicative given the six-positive training constraint; the shape function instability itself (§4.7, §5.2) remains the more direct, independent evidence for structural non-stationarity.

Chapter 6: Conclusions

6.1 Summary of the Research

This dissertation investigated the structural conditions under which ML-based EWS fail to achieve genuinely prospective predictive performance, and documented an instability mechanism consistent with the breakdown of their generalisation across crisis periods. The research addresses a gap identified by Beutel, List and von Schweinitz (2019): while machine learning outperforms logistic regression within expanding-window designs, the mechanism by which these improvements break down under genuinely prospective validation remained incompletely understood.

The EBM benchmark and ensemble extension, introduced beyond the original three hypotheses, yield two further results already established in Chapters 4 and 5 — shape-function training-window instability and ensemble probability accuracy — which are restated here as conclusions in Sections 6.2.3 and 6.2.5 rather than newly argued.

This chapter draws together the principal conclusions and the MSc AI contribution and closes with a final reflection; consistent with the limitations and their mitigations set out in §5.7, each conclusion is stated at the level of confidence the evidence supports — regime-specific where the scored positives concentrate in the single 2007 episode, and prospective where the strict temporal splits provide the binding test.

6.2 Main and Supplementary Conclusions

6.2.1 Extending Training History Does Not Resolve the Generalisation Problem

The first and most important conclusion is that the prospective failure of macro-based EWS is not an artefact of insufficient training data. M2 trained on the extended panel’s full pre-GFC history (1989–2002) achieves AUROC 0.481 in Split B strict temporal validation, below the random classifier baseline, providing evidence that fifteen additional years of crisis history including the Nordic cluster and Japan 1997 do not resolve the GFC generalisation problem. The results indicate that the additional pre-GFC training history available in this panel was insufficient to bridge the observed regime discontinuity between pre-GFC crisis types and the GFC.

The gap between expanding-window and strict temporal validation illustrates why the training-volume explanation fails: expanding-window evaluation progressively absorbs observations from later crisis regimes into the training sample before testing on subsequent folds, which inflates performance estimates for reasons unrelated to how much pre-crisis history is available. Split B removes that inflation, and the collapse to an AUROC of 0.481 shows that the additional training years contribute no genuine prospective benefit once the bias is stripped out.

6.2.2 The Interpretability-Accuracy Trade-off Is Not Binary

M2.5 (EBM) exceeds M2 under the Stage 1 expanding-window design, while M2.5-matched exceeds M3 within the separate 2007 cross-sectional case study, in both cases providing natively interpretable shape functions, which demonstrates that the interpretability-accuracy trade-off is not binary within the non-linear model class under standard evaluation conditions. Under strict temporal validation, EBM achieves AUROC 0.333 against RF's 0.481 in Split B; both fail the 0.75 benchmark, so the primary conclusion is that neither algorithm generalises prospectively regardless of interpretability. The 0.148-point gap is directionally consistent with higher flexibility incurring greater regime-specific transfer cost, but cannot be confirmed from six positive training cases — the shape function analysis (§4.7, §6.2.3) provides stronger, independent evidence for the same mechanism.

6.2.3 EBM Shape Functions Make Training-Window Instability Directly Visible

The dissertation's most theoretically significant contribution follows: revealing an interpretable mechanism consistent with prospective failure, namely instability in the fitted feature-risk mapping across training windows. Because EBM shape functions are the model itself rather than a post-hoc approximation, the instability observed across the two nested training windows cannot be attributed to attribution error. At `term_spread_lag3`, the highest-ranked EBM feature, the pre-GFC and GFC-inclusive shape curves cross, meaning the same feature value carries opposite-sign risk contributions across the two training windows. The pattern is consistent with DGP non-stationarity, though regime-specific model variance under the thin pre-GFC sample cannot be fully excluded (§4.7.3). The structural break test for credit growth (§4.9.7) is consistent with the DGP interpretation; the panel logistic sign reversal (§4.9.5) reflects non-monotonic lag structure rather than direct evidence of temporal change; the term spread Chow test falls short of significance, possibly reflecting limited power, although genuine coefficient stability cannot be excluded. Subject to these qualifications, the shape function instability is the dissertation's most direct evidence that static, pooled EWS designs face a structural limitation in non-stationary environments, made directly visible rather than inferred from metric comparisons alone.

6.2.4 Sentiment Augmentation Contributes a Directional but Unstable Signal

A further conclusion is that FinBERT sentiment does not contribute a robust or generalisable predictive signal. Within the 2007 GFC cross-sectional country ranking, M3 shows a directional AUPRC advantage over M2-matched (0.411 versus 0.343); however, this comparison confounds algorithm and feature changes simultaneously, the confidence interval includes zero, no auxiliary sentiment test survives Benjamini–Hochberg correction (0 of 156

Granger tests, §4.9.4; panel logistic non-significant, §4.9.6), and the isolated algorithm-controlled comparison (§4.2.1) shows sentiment reducing rather than improving EBM's discrimination within the same episode (-0.078 AUPRC, bootstrap 95% CI $[-0.292, +0.187]$). Sentiment is best characterised as a sample-specific correlation within the 2007 episode that does not survive algorithm control or auxiliary statistical testing, and cannot be assessed under multi-episode validation within the available sentiment window. Both auxiliary analyses are consistent with macro features carrying the dominant measured signal in this sample: SHAP attribution (73.8 per cent of M3's aggregate attribution) and the panel logistic pseudo- R^2 increment (0.034, non-significant at $p = 0.103$).

6.2.5 Supplementary Result: Ensemble Averaging Lowers Brier Score Within the 2007 Cross-Section

Finally, ensemble averaging lowers overall probability error within this same cross-sectional setting: E1-EBM records the lowest Brier Score of any case-study configuration (0.056), consistent with the Holopainen and Sarlin (2017) motivation for combining architecturally distinct models.

The gain is in overall probability accuracy and AUROC rather than AUPRC, and remains conditioned on the single GFC episode.

6.3 Contribution to the MSc in Artificial Intelligence

The project contributes to MSc AI learning outcomes across four dimensions.

In technical ML implementation, the pipeline spans XGBoost, Random Forest, EBM gradient-boosted GAMs, and FinBERT transformer-based sentiment inference, with calibrated probability outputs, bootstrap CIs, and an ensemble extension, demonstrating applied competence across supervised learning, NLP, and model evaluation.

In critical evaluation of AI systems, the project applies strict temporal validation and distribution shift analysis to reconcile high in-sample performance with prospective collapse, demonstrating that expanding-window metrics are systematically optimistic under regime non-stationarity; the honest reporting of negative results (below-random prospective AUROC, a null sentiment extension, a null topic-disaggregation result) reflects professional standards often absent from applied ML studies.

In research methodology and academic communication, the analysis integrates expanding-window cross-validation, Granger causality, panel logistic regression, SHAP attribution, and EBM shape function analysis within a coherent framework, with results reported against pre-specified benchmarks and explicit uncertainty quantification.

In ethical and professional responsibility, the project uses publicly available secondary data, operates under low-risk ethics clearance, fixes random seeds throughout, documents the full software environment, and makes code and outputs publicly accessible at <https://github.com/spike2025-art/eportfolio-uoe/tree/main> and the ePortfolio at <https://spike2025-art.github.io/eportfolio-uoe/>.

6.4 Final Reflection

This dissertation began with a model comparison question: does sentiment augmentation improve crisis prediction? It ended with a theoretical contribution about the structural limits of predictive modelling under distribution shift.

The most important lesson from the analysis concerns the relationship between machine learning and non-stationarity, not any single model, feature, or metric. A model trained on historical data represents that history; when the process that generated it changes, that representation becomes a misspecification. EBM's contribution is to make this visible rather than inferred: because its shape functions are the model itself, not a post-hoc approximation, the training-window-dependent instability documented in §4.7 and §5.2, supplemented by the exploratory cross-regime evidence in Appendix A, is a direct, readable property of what the model has learned, not an artefact of the attribution method.

That visibility does not resolve the non-stationarity problem, but it is a precondition for addressing it: a practitioner who can see where a model's learned mapping breaks down can apply appropriate scepticism to its outputs, communicate that limitation to policy audiences, and design the regime-conditional extensions proposed in §5.8. The contribution of interpretable machine learning in this setting is therefore not superior accuracy (Split B shows EBM's prospective performance is no better than Random Forest's, §4.5.1) but superior legibility: knowing not only what a model predicts, but the conditions under which that prediction should not be trusted.

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Appendix A: Exploratory Regime-Conditional Prototype

A.1 Scope and Purpose

This appendix presents an exploratory proof-of-concept implementation of regime-conditional modelling, as proposed in Section 5.8 (Proposed Extension 1). It is included not to establish a superior forecasting model, but to demonstrate the methodological feasibility and limitations of operationalising regime-awareness within the existing EWS pipeline. Results should not be interpreted as confirmatory evidence of a validated predictive system.

The prototype uses two manually defined regimes derived from the broader pre-GFC/GFC distinction that motivates the shape function analysis in Section 4.7.3 — specifically, the pre-GFC period (1989–2002, covering the Nordic banking crisis cluster and the Japanese crisis of 1997) and the GFC-inclusive period (2003–2020). This manual definition avoids the additional model selection problem that a data-driven clustering approach would introduce (choice of algorithm, distance metric, and number of regimes), and connects the prototype directly to evidence already presented in the main dissertation. A separate EBM is trained on each regime using the same hyperparameter specification as the NB05B benchmark (Section 4.4), and results are compared against a pooled EBM trained on the full 1989–2020 panel.

The evaluation uses a held-out test set of 2006–2007, which captures the 24 *target_h2* forward-looking crisis labels corresponding to the GFC onset. This window was mechanically determined by the temporal location of the available GFC target labels: the *target_h2* construction (crisis in $t+1$ or $t+2$) means GFC crisis events ($\text{crisis}_{\text{tst}} = 1$ in 2008) appear as positive labels in 2006–2007, with no further crisis positives in the panel after that point. The test set therefore represents the only feasible prospective evaluation window available in the current data.

Note: EBM-R2 (Regime 2 GFC-specific model) is trained on 2003–2005 and has only 2 crisis positives. Its metrics are reported for completeness but must be treated as statistically unreliable. All substantive interpretation focuses on EBM-Pooled vs EBM-R1.

A.2 Regime Sample Statistics

Table A.1 reports the sample composition of each regime and the test set. The key observation is the scarcity of crisis positives per regime — 11 for Regime 1 and 26 for Regime 2 — which is the binding statistical constraint identified in Section 5.8. This constraint prevents reliable shape function estimation in thin-sample conditions and is the primary reason a fully validated regime-conditional pipeline falls outside the scope of the present study. The test set base rate of 0.667 is unusually high, reflecting the temporal concentration of GFC forward labels in the

2006–2007 window; this elevated baseline must be borne in mind when interpreting AUPRC values in Section A.3.

Regime	Period	Rows	Crisis positives (target_h2)	Base rate	Crisis types
Regime 1 — Pre-GFC	1989–2002	250	11	0.044	Nordic cluster; Japan 1997
Regime 2 — GFC-inclusive	2003–2020	269	26	0.097	GFC 2008; post-GFC
Full panel (pooled baseline)	1989–2020	569	37	0.065	All crisis types
Test set (OOS evaluation)	2006–2007	36	24	0.667	GFC forward labels (target_h2)

Table A.1. Regime split statistics. Target variable: target_h2 (2-year forward crisis indicator). Crisis types are indicative based on JST chronology. Test set base rate of 0.667 (highlighted) reflects the concentration of GFC forward labels in 2006–2007; all AUPRC values reported in Table A.2 must be assessed against this no-skill precision-recall baseline.

A.3 Prospective Performance Comparison

Table A.2 reports AUROC, AUPRC, and Brier Score for each model evaluated on the 2006–2007 test set. All models are trained exclusively on data prior to the test window, preserving strict temporal separation. The 0.75 AUROC benchmark follows the practical-use threshold adopted throughout the main dissertation (Beutel, List and von Schweinitz, 2019). The no-skill baseline row shows the random AUROC (0.500) and the test-set positive prevalence (0.667), which is the no-skill precision-recall baseline for this evaluation; AUPRC values below 0.667 indicate performance below the no-skill precision-recall baseline implied by the test-set positive prevalence.

Model	Training period	Train+	AUROC	AUPRC	Brier	Meets 0.75 benchmark?
EBM-Pooled	1989–2005	13	0.410	0.636	0.606	No
EBM-R1 (Pre-GFC regime)	1989–2002	11	0.410	0.617	0.664	No
EBM-R2 (GFC regime) †	2003–2005	2 †	0.392	0.644	0.296	Unreliable †

Model	Training period	Train+	AUROC	AUPRC	Brier	Meets 0.75 benchmark?
<i>No-skill baseline</i>	—	—	<i>0.500</i>	0.667	—	—

Table A.2. Prototype OOS performance — test set 2006–2007 (36 rows, 24 positives, base rate 0.667). † EBM-R2 train positives on 2 crisis positives; results are statistically unreliable and excluded from interpretation. Red AUPRC values fall below the 0.667 no-skill precision-recall baseline.

AUROC benchmark: 0.75 (Beutel, List and von Schweinitz, 2019).

A.3.1 Interpretation

Four observations follow from Table A.2. First, all models fail the 0.75 AUROC benchmark. This is consistent with the main dissertation finding that prospective performance collapses under strict temporal validation (Section 4.5, Split B: EBM AUROC = 0.333; RF AUROC = 0.481). The prototype does not resolve this failure.

Second, and more importantly, all three AUPRC values (0.636, 0.617, 0.644) fall below the no-skill precision-recall baseline of 0.667 implied by the test-set positive prevalence. This means that none of the three models exceeds the no-skill precision-recall baseline implied by the test-set prevalence. The elevated base rate of the 2006–2007 test window — a consequence of the temporal concentration of GFC forward labels — makes AUPRC a particularly stringent metric here; nonetheless, the result further confirms that prospective discrimination is not improved by regime-specific training under the current data constraints.

Third, EBM-Pooled and EBM-R1 produce identical AUROC (0.410). Training exclusively on pre-GFC crisis episodes provides no discrimination advantage over the pooled model when evaluated against the GFC test set. This supports the interpretation that the pre-GFC feature-risk mapping does not transfer well to the GFC regime — which is precisely the non-stationarity mechanism the dissertation identifies. The absence of improvement should not be read as a failure of the prototype design, but as evidence consistent with the dissertation’s core thesis.

Fourth, EBM-R1 achieves a higher Brier Score than EBM-Pooled (0.664 vs 0.606), indicating worse probability calibration when restricted to pre-GFC training data. Recall that for Brier Score, lower is better. The pooled model benefits from the larger and more recent training sample even when that sample spans structurally distinct regimes, suggesting that in thin-data conditions the quantity of training positives partially compensates for the regime mismatch in terms of probability calibration, even where AUROC discrimination is unchanged.

A.4 Feature Importance Comparison

Important note on model scope: The feature-importance and shape-function comparisons in Sections A.4 and A.5 are explanatory diagnostics fitted on the full specified regime samples (EBM-Pooled on 1989–2020; EBM-R1 on 1989–2002; EBM-R2 on 2003–2020). These full-sample diagnostic fits are used here because they produce more stable importance estimates and shape functions than the restricted training windows used in the OOS evaluation. They are not used for any prospective performance claim. Prospective evaluation is reported exclusively in Table A.2, using models trained only on data prior to the 2006–2007 test window.

Table A.3 and Figure A.1 compare the top-10 main-effect features by mean absolute log-odds contribution for EBM-Pooled and EBM-R1. EBM-R2 is shown in Figure A.1 for visual reference but excluded from Table A.3.

Rank	EBM-Pooled — diagnostic (1989–2020)	Importance	EBM-R1 Pre-GFC — diagnostic (1989–2002)	Importance
1	term_spread_lag3_dm	0.791	cpi_lag2_dm	0.352
2	term_spread_lag2_dm	0.432	cpi_lag1_dm	0.324
3	term_spread_lag1_dm	0.384	debtgdp_lag3_dm	0.310
4	debtgdp_lag3_dm	0.272	cpi_lag3_dm	0.213
5	tloans_gr_roll3_std_lag1_dm	0.218	tloans_gr_roll3_std_lag1_dm	0.203
6	hpnom_gr_lag1_dm	0.174	stir_lag3_dm	0.160
7	cpi_lag1_dm	0.145	term_spread_lag3_dm	0.147
8	money_gr_lag1_dm	0.131	tloans_gr_roll3_mean_lag1_dm	0.130
9	tloans_gr_lag1_dm	0.118	hpnom_lag3_dm	0.127
10	tmort_gr_lag1_dm	0.109	term_spread_lag2_dm	0.112

Table A.3. Top-10 EBM main-effect features by mean absolute log-odds contribution — full-sample diagnostic fits. Feature suffix `_dm` denotes expanding-window demeaned. Importance scores are not directly comparable across models trained on different samples; the directional difference in feature rankings is the relevant quantity.

A.4.1 Interpretation

The feature hierarchy shift between EBM-Pooled and EBM-R1 is the most substantively important finding of the prototype. In the pooled diagnostic model, all three lags of term spread dominate the importance ranking (lag3: 0.791, lag2: 0.432, lag1: 0.384), consistent with the NB05B results reported in Section 4.6. In EBM-R1, trained exclusively on pre-GFC observations, term spread is largely absent from the top features: it appears only at rank 7 (lag3: 0.147) and rank 10 (lag2: 0.112). Instead, the top three features are CPI lags (lag2: 0.352, lag1: 0.324) and government debt-to-GDP at lag 3 (0.310).

This divergence is economically interpretable. The Nordic banking crises of the early 1990s and the Japanese crisis of 1997 were preceded by episodes of high inflation, rapid credit expansion, and fiscal deterioration — a macro-financial profile in which CPI and debt dynamics carry genuine predictive signal. The GFC, by contrast, was preceded by term spread compression in an environment of low inflation, making term spread the dominant pre-crisis indicator for that regime. In the full-sample pooled diagnostic model, the GFC crisis observations account for the majority of positive cases and therefore dominate the learned feature ranking. The prototype *suggests* that this dominance is not merely an artefact of model specification but a structural property of the pooled training distribution: when trained exclusively on pre-GFC data, the EBM recovers a fundamentally different feature hierarchy.

Figure A.1 (feature importance bar chart) illustrates this divergence visually. Readers should note that importance magnitudes are not directly comparable across models trained on different sample sizes; the directional difference in feature rankings is the relevant quantity.

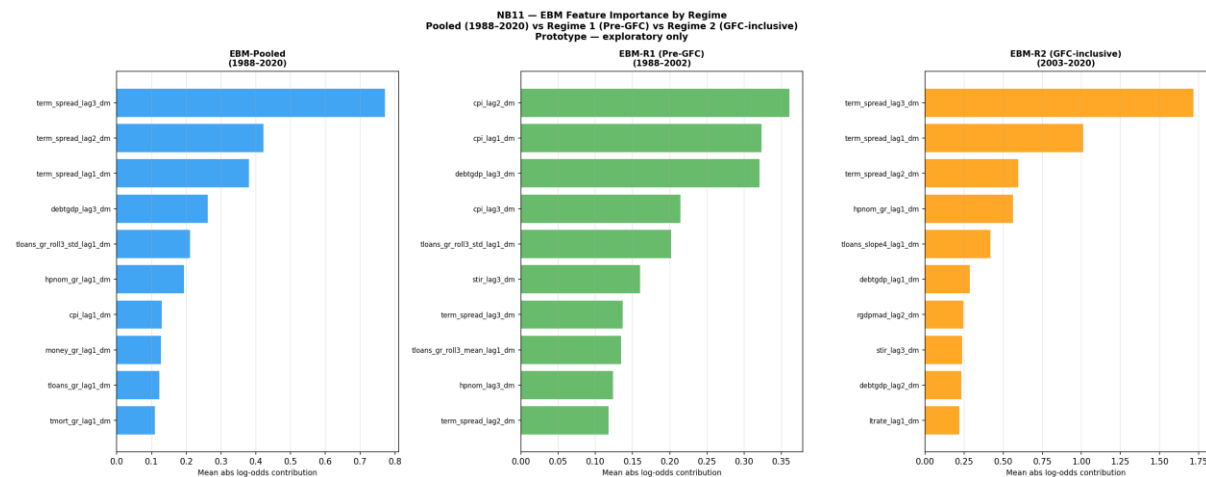


Figure A.1. EBM feature importance by regime — full-sample diagnostic fits. Left: EBM-Pooled (1989–2020), term spread dominates at all lags. Centre: EBM-R1 Pre-GFC (1989–2002), CPI and debt-to-GDP displace term spread as leading features. Right: EBM-R2 GFC-inclusive (2003–2020), term spread dominates consistent with GFC dynamics. The feature hierarchy shift between Pooled and R1 is the primary empirical finding of the prototype. Prototype — exploratory only.

A.5 Shape Function Analysis

Figure A.2 extends the shape function comparison of Figure 4.7 to three model configurations: EBM-Pooled, EBM-R1 (Pre-GFC), and EBM-R2 (GFC-inclusive). All three are full-sample diagnostic fits as defined in Section A.4. The three panels show the learned log-odds contribution of `term_spread_lag3_dm`, `term_spread_lag2_dm`, and `term_spread_lag1_dm` across the demeaned feature value range.

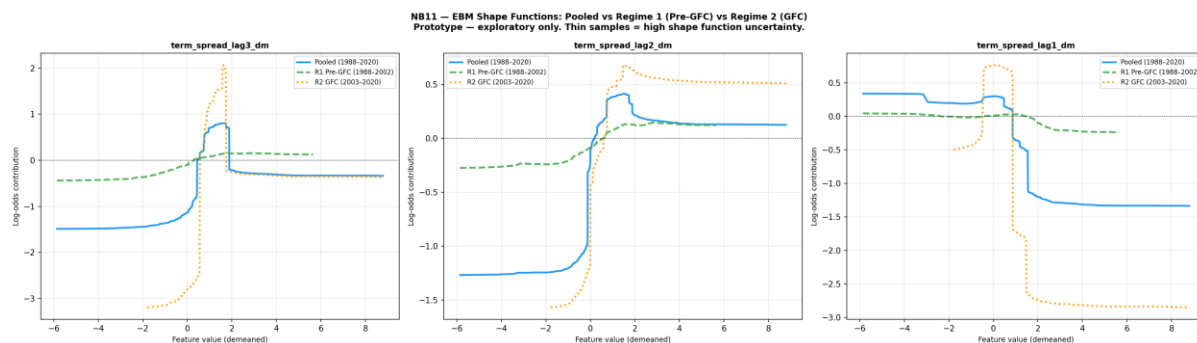


Figure A.2. EBM shape functions for term spread (lags 1–3): Pooled (blue solid), R1 Pre-GFC (green dashed), R2 GFC-inclusive (orange dotted). All three lags show near-zero R1 curves and high-amplitude R2 curves, consistent with term spread carrying no crisis signal in the pre-GFC regime and extreme non-linear signal in the GFC regime. Full-sample diagnostic fits — see Section A.4 note on model scope. Prototype — exploratory only; thin regime samples imply high shape function uncertainty.

A.5.1 Interpretation

The shape function divergence across the three configurations is the strongest visual evidence produced by the prototype. For all three lags of term spread, the EBM-R1 curve (green dashed) is near-flat across the full feature value range, with log-odds contributions close to zero. Term spread carries effectively no crisis-predictive signal in the pre-GFC regime. The EBM-R2 curve (orange dotted) shows extreme amplitude: for `term_spread_lag1_dm`, the contribution swings from approximately +0.5 to −3.0 log-odds, a 3.5-unit amplitude. The pooled model (blue solid) inherits a shape intermediate between these two extremes, dominated by the GFC signal because that regime contributes the majority of positive cases in the full-sample pooled fit.

This pattern directly extends the finding of Figure 4.7, where the pre-GFC and GFC-inclusive EBM curves diverge for term spread. The prototype adds a third configuration and supports the interpretation that the near-zero pre-GFC sensitivity is not an artefact of comparing two overlapping training windows: it is also observed in the EBM trained exclusively on pre-GFC data. The shape function instability is not symmetric — the pre-GFC regime is insensitive to term spread, while the GFC regime is extremely sensitive to it in a highly non-linear, threshold-driven manner. A pooled model cannot represent both structures simultaneously without misspecifying one or both.

A.6 Conclusion

The prototype produces three findings, each consistent with and supportive of the dissertation’s core argument. First, all regime-specific models fail the 0.75 AUROC benchmark and all AUPRC values fall below the 0.667 no-skill precision-recall baseline on the 2006–2007 GFC test set; EBM-R1 achieves identical AUROC to EBM-Pooled (0.410), supporting the interpretation that pre-GFC training data does not provide a transferable crisis-prediction signal for the GFC regime.

Regime-conditional training does not resolve the prospective failure when the available crisis positives per regime are insufficient to reliably estimate the feature-risk mapping.

Second, the feature hierarchy shift between EBM-Pooled and EBM-R1 — from term spread dominance to CPI and debt dominance — provides suggestive empirical evidence that the crisis-generating mechanism differs structurally across regimes. The pooled model learns a distribution dominated by GFC-era observations that is likely to misrepresent the pre-GFC crisis signature.

Third, the shape function comparison shows that term spread carries near-zero predictive signal in the pre-GFC regime and extreme non-linear signal in the GFC regime; a pooled model cannot represent both structures simultaneously without misspecifying one or both. Together, these findings support the conclusion stated in Section 5.8: regime-conditional modelling is conceptually well motivated and empirically supported as a future research direction, but its implementation is constrained by the scarcity of crisis episodes per regime in the current panel. Resolving this constraint — through a longer historical panel or an alternative multi-country crisis database — remains the most impactful direction for future research.

Appendix B

Appendix B: Software and Reproducibility

This appendix provides a complete inventory of the Jupyter notebooks comprising the analytical pipeline used in this study. All notebooks were developed in Python 3.11.14 within a dedicated Conda environment (Dissertation) and are listed below in execution order. The pipeline proceeds from exploratory data preparation through NLP-based sentiment extraction, model training, robustness evaluation, ensemble construction, and the topic-sentiment extension reported in Chapter 5.

Key software dependencies:

Python 3.11.14 • Interpret 0.7.8 (EBM) • xgboost 3.2.0 • scikit-learn 1.8.0 • shap 0.50.0 • transformers / ProsusAI/finbert • statsmodels • pandas / numpy / matplotlib

Data: JST Macrohistory Database R6 — 18 advanced economies, 1985–2020. BIS speech corpus: 16,622 speeches. Key analytical file: df_macro_augmented.csv (569 rows, 101+ columns).

Repository: <https://github.com/spike2025-art/eportfolio-uoe> • ePortfolio: <https://spike2025-art.github.io/eportfolio-uoe/>

Table B.1: Canonical Pipeline Notebook Inventory

#	Notebook File	Pipeline Stage	Description
01	01_Exploratory_Analysis.ipynb	Data Preparation	Loads and inspects the JST Macrohistory Database R6. Performs coverage audit across 18 advanced economies (1985–2020), identifies missing values, and confirms the two-horizon crisis indicator (target_h2). Produces the final analytical panel of 569 country-year observations used in all subsequent stages.

#	Notebook File	Pipeline Stage	Description
02	02_FinBERT_Setup_Test.ipynb	NLP Initialisation	Validates the FinBERT environment (ProsusAI/finbert via HuggingFace Transformers). Confirms GPU/CPU availability, tokeniser behaviour, and probability output format (P_pos, P_neg, P_neu) prior to full corpus processing in Notebook 03.
03	03_BIS_Sentiment_Extraction.ipynb	Sentiment Extraction	Processes 16,622 BIS central bank speeches through FinBERT (~12.4 hours runtime). Extracts sentence-level sentiment probabilities, maps speeches to JST country codes via central bank name lookup, and exports an annotated speech-level sentiment file for aggregation.
04	04_Sentiment_Aggregation _corrected.ipynb	Feature Engineering	Aggregates speech-level FinBERT scores to annual country-level sentiment features. Constructs rolling-demeaned P_neg series and generates lag-1 to lag-3 variants for integration into the macro panel. Produces the key output file df_macro_augmented.csv (569 rows, 101+ columns).

#	Notebook File	Pipeline Stage	Description
05	NB05A_Augmented_Model_Training.ipynb	Model Training (Primary)	Constructs the augmented macro panel (1989–2020) from JST R6 raw data. Builds all 41 macro features including level lags (lag 1–3), five growth rates, three rolling credit measures (3-year rolling mean, 3-year rolling standard deviation, 4-year OLS slope of total loans), and the term spread. Applies expanding-mean demeaning (country-specific, prior years only) to prevent forward leakage. Merges FinBERT sentiment features from jst_sentiment_master.csv and applies 3-year rolling demeaning; a substring-match pattern in the feature-selection step (Cell 12) yields 20 sentiment features rather than the 14 originally intended, so M3 trains on 61 features in total (41 macro + 20 sentiment) — a feature-selection labelling discrepancy, not a validity issue (see §3.3). Defines Stage 1 dataset (df_macro, 37 target_h2 positives) and Stage 2 dataset (df_sent, 26 positives, 2003–2020). Trains M1 (Logistic Regression, lag1 only), M1b (bridge LR, lag1–3), M2 (Random Forest, lag1–3), M2-matched (RF restricted to 2003–2020), and M3 (XGBoost + FinBERT, isotonic calibration). Exports out-of-sample prediction CSVs and feature lists consumed by NB05B and NB06B.
	NB05B_EBM_Interpretability_Benchmark.ipynb	Model Training & Interpretability	Trains M2.5/EBM (Explainable Boosting Machine) on the Stage 1 (1989–2020) and Stage 2 (2003–2020) datasets using expanding-window cross-validation. Produces regime-comparison shape functions for term spread and credit growth (early GFC-run-up 2003–2006 vs GFC-inclusive 2003–2010) as evidence of fitted-model and training-window instability consistent with possible data-generating-process instability. Exports EBM OOS prediction CSVs and feature importance rankings consumed by NB06B.

#	Notebook File	Pipeline Stage	Description
06	NB06B_EBM_Results_Analysis.ipynb	Stage 1 & 2 Evaluation	Computes Stage 1 discrimination metrics (AUROC, AUPRC) across all models: EBM AUROC=0.753, RF=0.636. Performs cross-sectional 2007 case-study evaluation (Stage 2): EBM AUROC=0.960, AUPRC=0.532; M3 AUPRC=0.411 vs M2-matched=0.343. Generates calibration curves, reliability diagrams, and all figures and tables reported in Chapter 4.
07	NB07B_EBM_Granger_Causality.ipynb	Causality Analysis	Performs Granger causality tests for credit growth and term spread on country-level sub-panels. Fits logistic regression of crisis indicator on term spread lags, reporting pseudo- $R^2=0.135$ and sign reversal between lag-1 (-0.646 , $p=0.004$) and lag-3 ($+0.630$, $p=0.003$), confirming non-monotonic predictor structure. Sentiment pseudo- R^2 increment: $+0.034$ ($p=0.103$, n.s.).
08	NB08B_EBM_Robustness_Checks.ipynb	Robustness	Implements three robustness exercises: (1) Leave-One-Country-Out (LOCO) evaluation across 15 economies (EBM mean AUROC=0.8931, std=0.2152); (2) prospective temporal splits — Split A Pre-Nordic (EBM AUROC=0.752, meets benchmark) and Split B Pre-GFC (EBM AUROC=0.333, RF=0.481, both below 0.75 benchmark); (3) ECE calibration analysis (EBM ECE=0.084).
09	NB09B_EBM_Ensemble_Extension.ipynb	Ensemble	Constructs ensemble E1-EBM by probability-averaging EBM and M2-matched predictions per Holopainen & Sarlin (2017). Reports E1-EBM AUROC=0.9545, AUPRC=0.4329, Brier=0.0557. Produces the 2007 country probability ranking (Figure 4.11) and crisis-type disaggregation (Fundamental, Contagion, No crisis). Generates model disagreement statistics reported in Chapter 4 §4.10.

#	Notebook File	Pipeline Stage	Description
10	NB10C_Topic_Sentiment _Models.ipynb	Extension (Ch. 5 §5.8)	Standalone topic-sentiment extension (completed study). Trains EBM with nine topic-sentiment features across three domains (credit, monetary policy, corporate), constructed from keyword taxonomy and FinBERT scores at lag-1 to lag-3. Core result: $\Delta\text{AUROC}=+0.000$, $\Delta\text{AUPRC}=+0.000$ — null finding consistent with macro-signal dominance within the evaluated sample (macro-to-topic importance ratio=0.240). Reported as Completed Extension in Chapter 5.
11	NB11_Regime_Conditional _Prototype.ipynb	Appendix A (§5.2, §5.8)	Implements the manually defined regime-conditional prototype: two temporal regimes — Regime 1, pre-GFC (1988–2002), and Regime 2, GFC-inclusive (2003–2020) — with pooled and regime-specific EBM sub-models. Produces the results reported in Appendix A (Tables A.1–A.3, Figures A.1–A.2) and summarised in §5.2 and §5.8. Exploratory proof-of-concept only.

NB10C is the final notebook of a three-notebook topic-sentiment sub-pipeline (NB10A: keyword taxonomy; NB10B: sentiment aggregation; NB10C: model training and evaluation). NB10A and NB10B are preparatory and do not produce reportable results independently. NB11 implements a proof-of-concept prototype for Proposed Extension 1 (§5.8) and is not referenced in the main empirical chapters. All figures and tables in Chapters 3–5 are produced by Notebooks 05A–09B; the Appendix A regime-conditional prototype is produced by NB11.

B.2 Supplementary Robustness Check: EBM Hyperparameter Sensitivity

To assess whether the baseline EBM configuration (5,000 rounds, 5 interaction terms, learning rate 0.01) reported in Chapter 4 was favourably selected, five alternative configurations varying the number of interaction terms (3, 5, 8) and the learning rate (0.005, 0.01, 0.05) were fitted on the same Stage 2 macro-only (41-feature) specification. AUROC was essentially flat across all five configurations (range 0.0071). AUPRC ranged more widely (0.1051), but the baseline’s own bootstrap 95 per cent CI [0.265, 0.827] contained every alternative configuration’s point estimate, meaning none was statistically distinguishable from the baseline. The baseline Brier score (0.0672) also outperformed three of the four alternative configurations. Taken together, the baseline configuration sits within the conservative range of reasonable hyperparameter choices rather than reflecting a search for the most favourable settings (referenced from Chapter 4, §4.7.4).

B.3 Supplementary Methodological Notes

Calibration mechanics (§4.8). A direct isotonic parity fix was not reported because the saved per-fold prediction files do not preserve the training splits, so it would be fit and evaluated on the same sample and drive ECE toward zero as an artefact. The gap between the EBM's lower ECE and its higher Brier Score reflects local versus global accuracy: ECE rewards bin-level alignment with the empirical crisis rate, while the Brier Score benefits from M3's within-fold isotonic calibration, so the EBM attains better bin-level alignment without that global squared-error optimisation.

Multiple-testing mechanics (§4.9.1). The Bonferroni correction controls the family-wise error rate by setting the threshold at $\alpha/m = 0.05/84 \approx 0.000595$. The Benjamini–Hochberg procedure controls the false discovery rate: it orders the 84 p-values and retains all tests up to the largest rank k for which $p(k) \leq (k/m)q$ at $q = 0.05$; here the largest qualifying rank is $k = 1$, so only the United States credit-growth result ($0.000015 \leq 0.000595$) is retained (Table 4.9a).